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The relationship between obesity and chronic kidney disease: Conclusions from a Kidney Disease: Improving Global Outcomes (KDIGO) Controversies Conference

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ABSTRACT

Obesity and chronic kidney disease (CKD) are complex diseases that are interlinked directly and indirectly. In October 2024, KDIGO (Kidney Disease: Improving Global Outcomes) held a Controversies Conference on Obesity and CKD to examine the most recent evidence regarding the epidemiology, pathophysiology, and treatment of obesity and CKD as well as to articulate priorities for research. A key conference theme was increasing awareness of obesity-related CKD. Long-term, early onset obesity as well as prolonged exposure to obesity carry the greatest risk of developing CKD. Identifying early biomarkers of kidney dysfunction and refining assessment methods in the context of obesity could provide opportunities for preventing loss of kidney function. The foundation for managing obesity in CKD comprises modifications to diet, physical activity level, and behaviors related to both. However, these strategies can be unsuccessful in achieving or maintaining weight loss for a myriad of reasons. Pharmacotherapies such as those including glucagon-like peptide-1 receptor agonists are effective in weight reduction and have been shown to have kidney-protective and cardiovascular benefits. Metabolic and bariatric surgery has also demonstrated benefits in reducing obesity-related complications. Appropriateness and choice of management strategies will vary depending on age and comorbidities and may change over time. Patient-lead goal setting is a foundation for dietary and physical activity interventions focusing on incremental, achievable changes towards healthy eating and an active lifestyle. Healthcare professionals require training to deliver these interventions and provide ongoing support with positive messaging using non-judgmental, stigma-free language. Evaluating the optimal duration of therapy, long-term safety of novel pharmacotherapies, and therapies in the context of early and later stages of CKD are key priorities for research. Multidisciplinary collaboration is important both for optimizing patient care and advancing research.

INTRODUCTION

Obesity is a complex chronic disease with multiple contributing factors, including environment, genetics, and social determinants of health. The prevalence of obesity in both adults and children is increasing, and it is expected that by 2035 more than 1.5 billion people will live with obesity.[1] In parallel, chronic kidney disease (CKD), which affects more than 800 million people globally, is also on the rise, both in prevalence and as a leading cause of mortality.[2, 3] Obesity is a driver for development of type 2 diabetes (T2D) and hypertension, the two leading causes of CKD in most countries.[4] In addition, obesity *per se* may lead to CKD directly.[5, 6] Adipose tissue may also affect the kidneys through production of adipokines, hormones that regulate inflammation and glucose metabolism, as well as through dysfunctional lipid metabolism, and each of these processes can contribute to CKD.[4]

Treating obesity with optimized diet and physical activity is preferred but often unsuccessful for a myriad of reasons.[7, 8] For example, reductions in weight are often not sustained. In recent years, trials evaluating metabolic and bariatric surgery (MBS) or pharmacological interventions have demonstrated positive outcomes,[9-11] with substantial reductions in weight that are maintained after MBS or with continued pharmacotherapy. It has been suggested that interventions leading to weight loss can reduce progression of CKD, as evaluated either by proxy through reduction in albuminuria, or directly through preservation of kidney function (i.e., estimated glomerular filtration rate [eGFR]). The interpretation of eGFR changes in the context of changing weight is complicated, however, since the biomarker creatinine, used in eGFR equations, is impacted by muscle mass and diet, and weight loss inevitably results in loss of some lean body mass.[12, 13]

In October 2024, KDIGO held a Controversies Conference on the Relationship Between Obesity and CKD: Pathophysiology, Prognosis, and Management. Participants discussed current evidence and controversies regarding epidemiology, pathophysiology, and diagnosis as well as the optimal means for addressing obesity in CKD with respect to nutrition, physical activity, pharmacological treatments, and MBS to meet person-centered needs in diverse populations. The format of the conference involved topical plenary session presentations (available on the KDIGO website: https://kdigo.org/conferences/ckd-obesity/) followed by focused discussion groups. Participants included 51 healthcare professionals and 3 patients. Key clinical issues and priorities for research were identified (Table 1).

EPIDEMIOLOGY OF OBESITY AND CKD AND ASSOCIATED RISKS

Current evidence shows that obesity, a chronic disease, has reached pandemic proportions. Since 1990, the prevalence of adult obesity worldwide has more than doubled, and adolescent obesity has quadrupled. In 2022, 2.5 billion people (43% of the population over 18 years) were overweight, and 1 in 8 people were living with obesity.[14] Similarly, 390 million children and adolescents aged 5–19 years were overweight in 2022, including 160 million living with obesity.[15] Thirty-seven million children under the age of 5 years were overweight.

The prevalences of both overweight and obesity generally increase with age, reaching a peak between the ages of 50 to 65 years and thereafter showing a slight downward trend. The prevalence of obesity is higher in women than in men of any age.[16] The age-standardized prevalence of obesity in adults has increased in 188 countries (94%) for women and in all except one country for men, but with distinct differences across countries.[17]

The long-term effects of obesity on the risk of CKD and adverse cardiovascular outcomes are multifaceted and well-documented in both adults and children.[5] The strongest evidence for this link comes from studies with long observation periods (up to 30 years) in children or young adults with obesity, demonstrating a clear connection to kidney disease[18] and adverse cardiac function.[19]

A high body mass index (BMI) is a well-recognized and strong independent predictor of the risk of CKD and kidney failure, even after adjustments for baseline blood pressure level and presence or absence of T2D.[20] A graded association has been shown between the degree of baseline BMI and the risk of kidney failure in a global meta-analysis of over five million adults (including representation from multiple Asian countries),[21] as well as in a large community-based sample of individuals in the United States.[20] In a study of a community-based cohort of 2,585 men and women from the Framingham Offspring Study with long-term follow-up (mean 18.5 years), each unit increase in BMI was associated with a 1.23-fold increased risk for new-onset kidney disease (as assessed by the Modification of Diet in Renal Disease Study equation and defined as GFR $\leq$ 59.3 ml/min per 1.73 m^2 for women and $\leq$ 64.3 for men), with the lowest risk for those with a BMI in the normal range and a higher risk for those at either high or low extremes of BMI.[22] For each BMI category, obesity in women was associated with higher health risks than in men (RR =1.92 vs 1.49), including a higher risk of kidney cancers (pooled RR 1.87 vs 1.53).[23]

Obesity affects 1 in 5 children worldwide.[24] A recent meta-analysis of 2033 studies in 45 million children from 154 different countries or regions showed that the prevalence of obesity was 8.5% and overweight was 14.8%, with distinct regional differences.[24] Since 1980 the prevalence of obesity has continuously increased and doubled in over 70 countries, with the rate of increase in childhood obesity greater than the rate of increase in adult obesity.[25] Lifecourse trajectories of BMI suggest that nearly 50% of children are on an increasing BMI trajectory with < 5% on a decreasing trajectory.[26]

The increasing prevalence of obesity in children raises the spectre of increasing prevalence of CKD in the future. Importantly, there is evidence that prolonged exposure to overweight during adult life increases the risk of later CKD in a cumulative manner.[27] A population-based, prospective study showed that people with overweight beginning early in adulthood (by ages 26 or 36 years) had approximately double the odds of developing CKD by age 60–64 years compared with those who first became overweight at age 60–64 years or never became overweight, with the strength of this association decreasing with increasing age of first overweight.[27]

At a population level, the prevalence of obesity is higher in populations with CKD G5 (eGFR <15 ml/min per 1.73m^2) versus age-matched general populations, showing the association between obesity and kidney failure. An approximately two-fold higher increase in BMI was reported in patients with incident kidney failure compared with the age-matched US population.[28] Controversy remains as to the extent and nature of this association, however, as some studies suggest that BMI has a U-shaped or J-shaped association with CKD progression, with the lowest incidence observed in people with overweight and mild obesity.[29, 30]

In exploring the causes of the obesity epidemic and risks for development of CKD, environmental and societal factors are potentially important. Some have suggested that the exposome—which includes factors such as global warming, air pollution, health disparities, microplastics, and lack of clean water—exacerbates the synergistic interactions of environmental factors with chronic diseases such as obesity and CKD.[31-33] Future research on the impacts of strategies combining policy and clinical care pathways may increase knowledge on developing effective interventions to prevent the growing population burden of obesity-related CKD.

PATHOPHYSIOLOGY LINKING OBESITY AND CKD

The progression from metabolic disruptions to development of obesity, CKD, and cardiovascular events and reduced life expectancy can be conceptualized as a "metabolic domino effect" (Figure 1). The complete details of this pathophysiologic progression are an important area of future study to identify areas for targeted interventions and preventive measures. T2D and blood pressure account for most of the risk associated with adiposity and CKD.[34] Chronic hypoxia has been proposed as the primary pathophysiological pathway driving CKD caused by T2D or other etiologies. Glomerular hyperfiltration, mitochondrial exhaustion, and a mismatch between oxygen delivery and oxygen demand may contribute to a cycle of hypoxic injury, glomerular loss, and progressive nephropathy (Figure 2).[35] Other metabolic alterations include inflammation, activation of the renin–angiotensin–aldosterone system (RAAS), oxidative stress, dyslipidemia, insulin resistance, lipotoxicity, gut dysbiosis, changes in adipokines, sodium and fluid retention, and the influence of obesogenic drugs. Adipose tissue accumulation in the kidney and liver positively correlates with increasing BMI, urine albumin-to-creatinine ratio (UACR), and triglycerides and negatively correlates with eGFR.[36] In addition to metabolic effects, the adipose tissue may also compromise kidney function by physical compression of the kidneys. Understanding these mechanisms is crucial for developing targeted strategies to mitigate CKD risk in patients with obesity and improve overall kidney health. [37]

The causal relationship between systemic inflammation and CKD is still being clarified. Ongoing trials investigating the effects of targeted IL-6 inhibitors, such as clazakizumab[38] and ziltivekimab,[39] in people with CKD with or without obesity may inform whether inflammation has a causal role in disease progression. Systemic inflammation is known to play an important role as a modifier of the paradoxical association between higher BMI and lower mortality in people undergoing hemodialysis.[40]

Genetic factors also influence the association between adiposity and CKD. Modeling based on Mendelian randomization analysis of UK Biobank data showed an effect of selected loci on the association of both general and central adiposity with CKD, although genetics appeared to be a weaker mediator of adiposity and CKD than diabetes or blood pressure.[34] Studies of genetic and environmental interactions will further our understanding of these complex relationships.

EVALUATION OF ADIPOSITY AND ASSESSMENT OF KIDNEY FUNCTION IN PEOPLE WITH OBESITY

Anthropometric measures of obesity

Despite the substantial evidence of an increasing prevalence of obesity and the importance of its associations with CKD, there is currently no gold standard assessment for obesity among people with CKD. BMI is the most widely utilized and available tool; however, relying solely on BMI can be misleading, and BMI cut-off values are not universal, with BMI >25 kg/m^2 used among Asians versus >30 kg/m^2 for others.[41] There is evidence that distribution of body fat, not obesity as defined by BMI *per se*, is related to a variety of physiological aberrations, such as hyperinsulinemia, hypertension, dyslipidemia, and atherosclerosis. Moreover, the risk of eGFR decline has been correlated with measures of central adiposity and the cardiovascular–kidney–metabolic (CKM) syndrome more so than with BMI.[42] Individuals with central fat distribution, regardless of BMI, have a demonstrated greater risk for diminished eGFR and albuminuria, with a graded association with increasing waist-hip ratio.[43] Conversely, in a population of 1,261 middle-aged persons (median age 59 years) without T2D, cardiovascular disease, or kidney disease, the presence of metabolic syndrome was an independent risk factor for accelerated age-related decline in iohexol GFR over a median of 5.6 years, with a 0.30 ml/min per 1.73 m^2 per year faster mean GFR decline, which was primarily driven by elevated triglyceride levels noted in metabolic syndrome.[44] A recently released consensus statement by the Lancet Commission on Definition and Diagnostic Criteria for Clinical Obesity[41] recommends the use of BMI only for population-level studies. It also recommends assessing excess adiposity at an individual level, either via direct measure of visceral organ adiposity or use of another anthropometric measure (such as WC, waist-to-hip ratio [WHR], or weight-to-height ratio), along with BMI. This statement differentiates preclinical and clinical obesity, which has therapeutic implications. The diagnosis of clinical obesity requires one or both of the following main criteria: 1) evidence of reduced organ or tissue function due to obesity (i.e., signs, symptoms, or diagnostic tests showing abnormalities in the function of one or more tissue or organ system) or 2) substantial, age-adjusted limitations of daily activities reflecting the specific effect of obesity on mobility, other basic activities of daily living (e.g., bathing, dressing, toileting, continence, and eating), or both. Those who fit the definition of clinical obesity would need urgent interventions to prevent adverse consequences. Future studies should assess the impact of using various combinations of anthropometrics on clinical outcomes in CKD.

Several measures have been used to study the adverse consequences of adiposity on kidney function or incident CKD and cardiovascular outcomes and mortality in those with pre-existing kidney disease.[45] The advantages and limitations of these measures for clinical practice are described in Table 2. WC and WHR were identified as risk factors for eGFR decline and death in individuals who have normal or reduced levels of eGFR. However, reported associations between BMI, WC, and kidney disease progression and mortality have been discordant,[46, 47] warranting evaluation of adiposity using body composition studies. An additional challenge of using anthropometric measurements is that high fluid retention and hypervolemia can distort not only body weight but also interfere with the accuracy and reliability of assessing obesity. For individuals, changes in BMI over time may not fully represent an increase in adipose tissue[48, 49] given the influence of multiple factors (such as fluid overload) on BMI in those with CKD.

Clinical phenotypes of obesity

Multiple body composition subtypes (visceral and subcutaneous obesity) and clinical phenotypes of obesity (e.g., sarcopenic obesity) have been identified in the general population. Even though differential risks of individual obesity subtypes with clinical outcomes have been reported,[50-53] there is currently no well-established diagnostic criterion or consensus on the use of appropriate diagnostic modalities for identifying such subtypes, and therefore further studies are warranted. Visceral obesity has been identified as an independent risk factor for CKD and associated with inflammation and adipokines in CKD.[51] However, radiological evaluation may be needed to evaluate visceral obesity in several situations, although this may not be universally feasible.[52, 54]

Sarcopenic obesity, a clinical condition characterized by the co-occurrence of excess adipose tissue and loss of skeletal muscle mass,[55] is associated with an increased risk of CKD and affects 2–23% of the CKD population.[53, 56] There is a lack of consensus on the definitions of sarcopenia and sarcopenic obesity, with considerable differences among diagnostic criteria, methodologies, and cut-off values. Nonetheless, the clinical consequences of sarcopenic obesity are considerably worse than those for either sarcopenia or obesity.[57]

Metabolically healthy obesity (MHO) refers to the presence of obesity in the absence of common metabolic risk factors. A large meta-analysis (n= 4,965,285 participants) illustrated a statistically significant association between risk of CKD and the phenotypes of being "metabolically healthy" overweight (RR: 1.29; 95% CI: 1.27–1.32, $P < 0.001$) or obese (RR: 1.47; 95% CI: 1.31–1.65, $P < 0.001$),[50] suggesting both preclinical and clinical obesity (in the absence of metabolic risk factors) is harmful, with additive effects for metabolic risk factors in addition to obesity.

Kidney function assessment in obesity

Accurate assessment of kidney function is often challenging in people with obesity and remains a subject of debate. Gold standard assessment tools including exogeneous glomerular filtration markers such as iothalamate, ethylenediaminetetraacetic acid, diethylene triamine pentaacetic acid, and iohexol, are relatively expensive, inconvenient, and time-consuming and are therefore rarely utilized in clinical practice.[58, 59] Serum creatinine and cystatin C, two endogenous molecules, are widely used to assess eGFR; however, their accuracy in GFR estimation is unclear among patients with extreme obesity (Figure 3).[60] Reliability of these measures in individuals undergoing weight loss also merits further studies.

There has been debate about the utility of indexing to body surface area (BSA), which has direct implications for CKD staging and drug dosing. There is a lack of validation of BSA-estimating formulas (non-indexed) in people with obesity, and indexing has the potential to mask glomerular hyperfiltration among people with obesity.[61] In an analysis of 3,611 participants from the CRIC (Chronic Renal Insufficiency Cohort) Study, increases in obesity-associated BSA were accompanied by higher absolute GFR for a given eGFR.[62] This was further evaluated in a study of 4,707 persons referred for measured GFR with a range of BMIs.[63] Analysis of indexed and non-indexed eGFR using creatinine, cystatin C and the combination concluded that indexed eGFRcr-cys was more accurate than indexed eGFRcr across the BMI spectrum. Using nonindexed eGFRcr-cys further improved accuracy for some treatment decisions. Studies comparing the effectiveness of indexed or unindexed GFR for the prediction of long-term CKD-related clinical outcomes including CKD progression, major adverse cardiovascular events, or mortality, are needed.

Albuminuria assessment in obesity

The presence of albuminuria is a well-studied and acknowledged risk factor for CKD. However, when assessing UACR in the setting of obesity, it is important to remember that urinary creatinine excretion is a direct reflection of muscle mass. This can be particularly problematic in sarcopenic obesity and in those undergoing weight loss.[64] A mild-to-moderate change in urinary creatinine excretion can lead to an under- or overestimation of spot UACR if urinary albumin remains constant.[64] Although determining a lower threshold for UACR may appear as a solution, such an approach may be misleading among people with sarcopenic obesity.

Issues Relating to Kidney Transplant

In people with kidney failure, including those living with obesity, kidney transplantation provides survival benefits over dialysis therapy.[65-67] However, obesity is associated with lower access to kidney transplantation as evidenced by lower rates for referral, longer wait times, and lower rates of transplantation.[68] Relative to individuals with a BMI of 18.5–25 kg/m^2, individuals with BMI ≥35 kg/m^2 are less likely to be listed for receiving a kidney transplant, especially among individuals <45 years old, women, or those of Asian race.[69]

The rationale for this diminished access is the documented higher risk for surgical complications (i.e., longer procedure period and warm ischemia time, higher risk for surgical-site infections, dehiscence, vascular complications including venous thromboembolism) and post-operative complications (i.e., delayed graft function, acute rejection).[70] However, whether these risks outweigh the known benefits of transplantation over dialysis remains somewhat controversial. Obesity treatment should be considered prior to transplantation, mediated via pharmacotherapy or MBS to facilitate timely access to kidney transplant and potentially improve long-term metabolic risks. However, more data in this area are needed. Multiple studies, mostly retrospective with small sample sizes and short follow-up periods, reported potential benefits of intentional weight loss prior to transplantation among kidney transplant candidates with obesity,[71-73] but larger-scale, prospective clinical studies investigating this highly relevant issue for clinical practice are needed. It is noted that after MBS, maintaining adequate immunosuppression levels after transplantation can be challenging.[74]

INTERVENTIONS FOR WEIGHT LOSS AND IMPROVING KIDNEY FUNCTION

Tailored diet and physical activity interventions should be a foundation of therapy in people living with obesity and CKD. Also integral to management are behavioral therapy and consideration of obesity-management medications and surgical interventions. The mechanisms by which weight loss affects kidney outcomes are unclear and may be direct or indirect. Weight change should be considered alongside changes in clinical and patient-reported outcomes, including kidney function, quality of life, blood pressure, functional capacity, and measures of diet quality.

Diet and Exercise

Randomized controlled trials (RCTs) have reported the effectiveness and safety of reduced energy intake and increased physical activity to achieve weight loss and improved functional capacity in people living with obesity and CKD.[7, 75] However, studies have not consistently shown evidence that behavioral interventions to reduce intake improve metabolic outcomes including blood pressure, albuminuria, and blood lipids, possibly due to small sample size, short follow-up, or study heterogeneity.[7] Studies addressing kidney outcomes in populations at high risk of CKD have shown that diet and physical activity interventions can reduce the incidence of reaching a high-risk CKD category (orange in the KDIGO heat map[76]) and slow eGFR decline.[77-80] Successful interventions, which include dietary modification and physical exercise, incorporate elements of behavioral therapy. In a secondary analysis of the LOOK-AHEAD (Action for Health in Diabetes) RCT, participants with T2D and obesity who received intensive lifestyle intervention achieved a 9% reduction in body weight versus 6% in the usual education group.[77] This was accompanied by a 31% reduced risk of incident very–high-risk CKD based on the KDIGO heat map (hazard ratio [HR] 0.69, 95% confidence interval [CI] 0.55-0.87).[76, 77] In a secondary analysis of the CORDIOPREV (CORonary Diet Intervention with Olive Oil and cardiovascular PREVention) RCT, participants with T2D and obesity randomized to a Mediterranean diet had less decline in eGFR than those randomly assigned to a low-fat diet, independent of weight loss.[79] In the PREDIMED-Plus (PREvención con DIeta MEDiterránea-Plus) RCT, participants with CKM syndrome randomly assigned to an energy-reduced Mediterranean diet along with physical activity recommendations and behavioral support had a 40% lower incidence of moderately or severely impaired eGFR (HR 0.60; 95% CI 0.44–0.82), and 92% higher reversion of moderately to mildly impaired eGFR (HR 1.92; 95% CI 1.35–2.73), when compared with a control group with education about a Mediterranean diet.[80]

Table 3 summarizes the key studies on dietary interventions for obesity management, including suggestions regarding suitability at different stages of CKD and evidence gaps in certain populations.[77, 80-88] Dietary and physical activity interventions focusing on incremental, achievable changes towards healthy eating and an active lifestyle with positive messaging and ongoing support can be implemented using patient-led goal setting.[86, 89] Healthcare professionals require training to deliver these interventions using non-judgmental, stigma-free language and approaches.[90]

Obesity Management Medications

Several medications are currently licensed with an indication for obesity treatment, and, of these, incretin mimetics have the most available data with respect to kidney outcomes. Incretin mimetics currently licensed or in development include glucagon-like peptide-1 receptor agonists (GLP-1 RA), glucose-dependent insulinotropic polypeptide receptor agonists (GIP RA), and combinations of these including triple combinations with glucagon receptor agonists and antagonists. At present GLP-1 RA and dual GLP-1/GIP RA have the most data to inform managing obesity in people with or without kidney diseases.[91] In people with CKD, these have shown efficacy for weight reduction with an absence of CKD-specific safety signals.[92-95] Of note, trials including participants with CKD excluded transplant recipients, and many excluded people with eGFR < 30 ml/min per 1.73 m^2. Dialysis and transplantation are not listed as contraindications to using GLP-1 RA or GLP-1 RA plus other peptides, although there is a lack of evidence of kidney benefits in patients with GFR < 15 ml/min per 1.73 m^2 (including those on dialysis). Emerging retrospective data indicate potential kidney benefits in kidney transplantation, tested mostly in transplant recipients with T2D,[96, 97] but RCTs are needed.

An important question is whether obesity management medications are, in practice, used long enough to benefit kidney health. Lack of adherence and non-persistence with treatment are important barriers;[98] in some real-world studies, non-adherence is as high as 60%.[99, 100] Important barriers to accessibility for GLP-1 RA are cost and payer preparedness.[101] There is a need for trial and real-world data in this population to inform strategies for realizing potential benefits of medication.

Effects of incretin-based therapies on the kidney

To date the only trial of incretin-based therapy with a kidney primary endpoint was the FLOW (Evaluate Renal Function with Semaglutide Once Weekly) trial. The FLOW trial examined semaglutide 1.0 mg weekly in people with T2D and CKD.[93] Prior transplant, kidney replacement therapy, or an eGFR < 25 ml/min per 1.73 m^2 were exclusion criteria. Overall, 400 of 3,533 total participants had eGFR < 30 ml/min per 1.73 m^2. The primary outcome comprised kidney failure (dialysis, transplantation, or eGFR < 15 ml/min per 1.73m^2), ≥ 50% eGFR reduction from baseline, or kidney or cardiovascular death. Relative to placebo, semaglutide was associated with an overall 24% reduction in the primary outcome and a 21% reduction (HR: 0.79; 95% CI: 0.66–0.94) in a composite outcome of kidney-specific components. There was no significant variation in this effect based on eGFR category at baseline. The mean annual eGFR slope was 1.2 ml/min per 1.73 m^2 less with semaglutide relative to placebo ($P < 0.001$). The benefits on slope were seen both for calculations using creatinine as well when using cystatin C to calculate eGFR.

The data from FLOW were consistent with prior secondary analyses from trials of semaglutide and other GLP-1 RA in people with T2D. A meta-analysis including data from seven trials reported a 21% reduction ($P < 0.001$) in the composite kidney outcome (Table 4). This composite consisted of development of severely increased albuminuria, doubling of serum creatinine, or either at least 40% decline in eGFR, kidney replacement therapy, or death due to kidney disease. Trial-specific effect estimates ranged from a 12% reduction of the composite outcome in EXSCEL (Exenatide Study of Cardiovascular Event Lowering) to a 36% reduction in SUSTAIN-6 (Trial to Evaluate Cardiovascular and Other Long-term Outcomes with Semaglutide in Subjects with Type 2 Diabetes), indicating a class effect.[92]

For incretin-based therapies, to date the largest study with kidney outcome data in the absence of T2D come from the SELECT (Semaglutide Effects on Cardiovascular Outcomes in People With Overweight or Obesity) trial. The SELECT trial included people with overweight and obesity and prior cardiovascular disease but without T2D at baseline.[94] Patients undergoing hemodialysis or peritoneal dialysis and individuals with eGFR < 15 ml/min per 1.73 m^2 were excluded. Among enrolled participants, 11% (1908/17,604) had an eGFR < 60 ml/min per 1.73 m^2 at study entry, with just 26 persons having an eGFR < 30 ml/min per 1.73 m^2. Semaglutide 2.4 mg subcutaneous weekly was associated with a 20% reduction in the primary endpoint of first occurrence of major adverse cardiovascular events (MACE) compared with placebo. The reduction in risk of MACE with semaglutide was similar in those with and without eGFR < 60 ml/min per 1.73 m^2 and in those with and without albuminuria at entry.[102]

For kidney outcomes in SELECT, semaglutide 2.4 mg subcutaneous weekly was associated with a 22% reduction in the prespecified composite kidney endpoint (HR: 0.78; 95% CI: 0.63–0.96; $P = 0.02$). The treatment difference of 2.19 ml/min per 1.73 m^2 in eGFR at 104 weeks of the trial was greatest in the 11% who had an eGFR < 60 ml/min per 1.73 m^2 at baseline (95% CI: 1.00–3.38; $P < 0.001$). While treatment discontinuations in both treatment arms were more common in patients with baseline eGFR < 60 ml/min per 1.73 m^2, a lower number of serious adverse events were reported in those taking semaglutide versus placebo.

There was no evidence of increased safety signals in more frail individuals or those with lower eGFR in FLOW and SELECT, respectively, though there were few persons with eGFR < 25 ml/min per 1.73 m^2 in the trials and none with kidney replacement therapy. In both trials there was a transient initial fall in eGFR with semaglutide. In SELECT this was a net 1.33 ml/min per 1.73 m^2 per year greater decline in patients randomized to semaglutide, but by week 20, eGFR was similar in both treatment arms overall. There was no increase in albuminuria during this period.

A recent meta-analysis of GLP-1 RA phase 3 and 4 trials[95] that included SELECT and FLOW reported an overall reduction across all trials of GLP-1 RA combined in the composite kidney outcome by 18% compared with placebo (HR 0.82, 95% CI 0.73-0.93; I^2 =26.41%), and a reduction in kidney failure of 16% (HR 0.84, 95% CI 0.72-0.99). The combined endpoint in that analysis excluded albuminuria and was restricted to kidney failure (i.e., kidney replacement therapy or a persistent eGFR < 15 ml/min per 1.73 m^2), a sustained reduction in eGFR from baseline by 50% or more, or death due to kidney disease.

Semaglutide was tested in a smaller study (n=101) dedicated to patients with overweight or obesity and chronic kidney disease without diabetes in a trial with albuminuria as primary endpoint.[103] Compared with the group receiving placebo, the group treated with semaglutide 2.4 mg had 52.1% lower UACR (95% confidence interval −65.5, −33.4; P < 0.0001) at 24 weeks.

Fewer data are available for tirzepatide, dual GLP-1 RA and GIP RA. A secondary analysis of the SURPASS (A Study of Tirzepatide [LY3298176]) Once a Week Versus Insulin Glargine Once a Day in Participants With Type 2 Diabetes)-4 open-label trial of tirzepatide versus insulin glargine in individuals with T2D reported a lesser fall in eGFR with tirzepatide on kidney outcomes (data pooled across three doses 5, 10, 15 mg).[104] Further analysis showed this was the case whether eGFR was estimated using measures of cystatin C or creatinine.[105] There was also an initial transient fall in eGFR with tirzepatide as seen with semaglutide. A recent pooled analysis across SURPASS trials 1-5 found no difference in eGFR between tirzepatide and pooled comparators at week 40/42.[106] However, reductions in UACR were 19%, 22%, and 26% for tirzepatide 5, 10, or 15 mg, respectively, compared with all pooled comparators. Longer-term data for tirzepatide will be available from the forthcoming SURPASS-CVOT (A Study of Tirzepatide [LY3298176] Compared With Dulaglutide on Major Cardiovascular Events in Participants With Type 2 Diabetes; NCT04255433) trial due to report in 2025.

Cotadutide combines GLP-1 RA and glucagon RA. In a small phase 2B study in people with T2D and CKD, cotadutide showed significant reductions in UACR relative to standard of care.[107]

Retatrutide is a triple combination of GLP-1 RA, GIP RA, and glucagon RA for which an analysis of 2 phase 2 trials—one in people with obesity and the other in those with T2D—has been done.[108] In the evaluation in participants with T2D, at 36 weeks there was no observed difference in Cr-eGFR between retatrutide and placebo, though UACR was significantly reduced with retatrutide. In the trials of participants with obesity, at 48 weeks retatrutide was associated with increased Cr-eGFR and decreased UACR compared with placebo.

For dipeptidyl peptidase-4 inhibitors, a systematic review found that only one study out of five trials with kidney data indicated a reduced risk of composite microvascular outcome and albuminuria progression.[109]

Mediation analyses of incretin mimetic therapies have noted that weight loss did not fully explain the kidney benefits. Thus, it is unclear if interventions such as medications benefit kidney health in the absence of weight loss and whether there should be a target weight or other outcome measure used in people with CKD.

Non-incretin medications used in obesity

Orlistat is a lipase inhibitor that reduces the absorption of dietary fat. A small, open-label study in people with CKD and BMI > 30 or > 28 kg/m^2 with comorbid conditions (T2D, hypertension, or dyslipidemia) found that the weight loss in the group that received orlistat together with individualized support and exercise prescription was associated with a slower decrease in eGFR than in the usual-care group (without weight management).[110] However, the authors noted that this result might have been affected by sampling bias. Additionally, concerns about acute kidney injury and oxalate nephropathy with orlistat have been raised,[111] leading to a label warning to monitor kidney function in patients at risk for kidney insufficiency and to discontinue the drug if oxalate nephropathy develops.

Phentermine is a Schedule IV controlled stimulant that is indicated for weight reduction in the United States, but for short-term use only, and it is not indicated in children. The label advises caution when used in people with kidney impairment. Naltrexone/bupropion is an opioid receptor antagonist also used for weight loss, although it is contraindicated in persons with kidney failure.

Other therapies with known kidney benefits but not licensed as obesity drugs

Sodium-glucose cotransporter-2 (SGLT2) inhibitors have beneficial effects on cardiovascular and kidney outcomes in people with and without CKD.[112] Metanalyses across all SGLT2 inhibitor trials suggest modest weight loss—a mean weight reduction of −1.79 kg (95% CI: −1.93 to −1.66, $P < 0.001$)—compared with placebo.[113] In several studies it has been noted that GLP-1 RA effects on kidney outcomes are independent of and show no interaction with background SGLT2 inhibitor use.[114]

Pediatric populations

In the United States, semaglutide, liraglutide, orlistat (120 mg), and phentermine–topiramate have been approved as obesity management medications for adolescents ≥ 12 years of age and weighing ≥ 60 kg with an initial BMI corresponding to 30 kg/m^2 or greater for adults. In contrast, only liraglutide and semaglutide have been approved for this age group in Europe. Data on the impact of obesity management medications on kidney outcomes in children are lacking. In the Semaglutide Treatment Effect in People with Obesity (STEP) TEENS trial, in adolescents with obesity or with overweight and at least one weight-related coexisting condition, semaglutide was associated with a net difference of −16.7 percentage points (95% CI −20.3 to −13.2; $P < 0.001$) in BMI. Data on eGFR or albuminuria were not reported.[115]

Surgical Interventions

Prior to the relatively recent introduction of incretin mimetic therapies, surgical interventions for obesity were one of the few therapies which had shown promise in yielding sustainable weight loss. The most commonly performed MBSs are sleeve gastrectomy or Roux-en-Y gastric bypass (RYGB). Overall, surgical weight loss has been shown to be both relatively safe and effective to induce weight loss in people with kidney disease and associated with improved kidney outcomes in individuals with eGFR < 60 ml/min per 1.73 m^2. Relative to RYGB, sleeve gastrectomy has advantages of not being associated with kidney stones or oxalate nephropathy and not interfering with absorption of immunosuppressive medications after kidney transplantation.[116]

Prior studies suggested that weight loss following MBS may decrease glomerular hyperfiltration as indicated by decreases in GFR (both estimated and measured) in people with eGFR > 120 ml/min per 1.73 m^2. However, the arbitrary definition of hyperfiltration as GFR > 120 ml/min per 1.73 m^2 can be problematic. In a retrospective, propensity-matched analysis comparing 985 people who underwent MBS with 985 who did not undergo surgery (33% of each group had baseline eGFR < 90 ml/min per 1.73 m^2), MBS was associated with a 58% lower risk for eGFR decline and a 57% lower risk of doubling of serum creatinine or kidney failure over a median follow-up of 3.8 years.[117]

CKD G5 and G5D. Patients with kidney failure on dialysis experience a higher incidence of post-MBS complications compared with those without kidney disease; however, the absolute complication rates are low relative to benefit and are not MBS-specific.[118] In a retrospective propensity matched analysis of 717,809 patients spanning 2015-2019, the cumulative risk of death within 30 days after MBS was approximately 0.6% in 5,817 CKD patients (versus 0.2% for those without CKD.[119] Current evidence suggests that MBS before kidney transplantation does not lead to adverse outcomes post-transplantation; however, robust evidence in this area is lacking.[120]

When MBS is performed, exercise training before and after is critical for preserving both muscle mass and fitness, as sarcopenia, and especially low muscle strength, are associated with a high risk of complications, including cardiovascular events and mortality, in patients on hemodialysis.[121, 122]

Pediatric populations. In many countries MBS is either not permitted or is permissible only in rare situations in children and adolescents with obesity. In the United States, MBS is recognized by the American Academy of Pediatrics as an effective treatment for severe obesity in childhood.[123] Sleeve gastrectomy is almost always the surgery type used in this population. Few, if any, randomized controlled trials exist in these populations. In observational data from the Teen-LABS (Longitudinal Assessment of Bariatric Surgery) cohort, MBS was associated with improved kidney outcomes in adolescents (aged 13 to 19 years) with severe obesity, including those with CKD (G1-G4) and T2D.[124] The procedure led to significant improvements in eGFR, resolution of albuminuria in some, and reduction of hyperfiltration. For adolescents with CKD G5/G5D or a kidney transplant, data on outcomes of MBS are lacking.

Comparing Drug Therapies and MBS

Figure 4 outlines the overall relative strengths and drawbacks of pharmacotherapy and MBS. Whether drug therapies or MBS are superior in achieving weight loss and safety endpoints has not been adequately studied; the current best-in-class FDA-approved agents (semaglutide and tirzepatide) have not been prospectively compared with MBS at their maximum tolerated doses for obesity management. Evidence from the SURMOUNT-1 (Tirzepatide Once Weekly for the Treatment of Obesity) study has indicated that the maximum dose of tirzepatide (15 mg) leads to comparable weight loss as MBS.[11, 125]

Few observational data exist directly comparing surgical and medical interventions. The M6 (Macrovascular and Microvascular Morbidity and Mortality after Metabolic surgery versus Medicines) was a retrospective observational study in the United States from 2010–2017 comparing nephroprotective effect of MBS with GLP-1 RA.[126] In that study, patients who underwent MBS had a significantly lower incidence of CKD progression (defined as the onset of ≥ 50% sustained decline in eGFR compared with baseline, development of sustained eGFR < 15 ml/min per 1.73 m^2, initiation of dialysis, or kidney transplant after the index date) after a median follow-up of 5.8 years. It is important to note, however, that none of the non-surgical patients was receiving best-in-class medications such as semaglutide and tirzepatide at baseline, and only 19% received these agents during follow-up, limiting the true comparison between medical and surgical management.

Uncontrolled studies comparing surgical and medical interventions using empaglifozin and liraglutide have shown comparable outcomes. The clinical study Metabolic Surgery Compared to the Best Clinical Treatment in Patients With Type 2 Diabetes Mellitus (MOMS) was a prospective, open-label, randomized study involving 100 patients with T2D, obesity, and CKD G1–G3a, A2–A3 (n=100) comparing RYGB versus best pharmacotherapy for patients with T2D and obesity (empagliflozin, liraglutide).[125] Albuminuria remission was not statistically different between pharmacotherapy and RYGB after 5 years in participants with diabetic kidney disease and class 1 obesity (BMI 30–35 kg/m^2). RYGB was superior in improving glycemia, diastolic blood pressure, lipids, body weight, and quality of life. However, more studies are needed that include the most effective GLP-1 RA prospectively escalated to the maximum tolerated dose to adequately compare medical and surgical therapies.

OPTIMAL MODELS OF CARE

Patient participants at the conference emphasized the importance of having multidisciplinary care for support in choosing therapy and adopting changes in nutrition and physical activity level. Indeed, effective interventions for obesity and CKD management require repeated interactions with healthcare teams, support for behavior change, flexibility, long-term monitoring, and follow-up, and they should be underpinned by behavior change models or theoretical frameworks.[91, 127] Individualized interventions must be supported by policy and advocacy at the community, national, and international levels to create long-term change. There is substantial ongoing controversy around models of care; approaches based on implementation science are needed to support the rapid translation of treatments into care for populations globally.

Optimal care models for people living with obesity and CKD should aim to integrate the best kidney care with the best obesity care by adopting a stigma-free, person-centered approach linking primary and secondary care, and involving a multidisciplinary team (MDT) trained to deliver long-term obesity and kidney care. Where possible, the MDT should include nephrologists, endocrinologists, obesity medicine physicians, cardiologists, kidney dietitians, bariatric dietitians, nurses, pharmacists, psychologists, and exercise professionals. Metabolic, kidney, and patient-reported outcomes (PROMs) including diet quality, quality of life, physical function, and patient-reported experience measures (PREMs) can provide valuable insights into physical and mental health and treatment satisfaction.[128, 129] People living with obesity and CKD should have access to holistic structured assessment, incorporating components of physical, mental, and socio-economic determinants of health.[128-130] Obesity is a key component in the CKM syndrome (it is recognized that the liver is connected to cardiovascular health and also affects metabolic functioning).[131, 132] Providing education on the CKM syndrome may enhance understanding and engagement in dual obesity and CKD management and support individualized goal setting.

Figure 5 depicts a multidimensional approach to obesity and CKD including acknowledgment of broader contexts and optimal integration of an MDT. Evidence has shown that person-centered care delivered by an appropriately trained MDT can improve CKM risk factors, PROMs, and PREMs, leading to reductions in risk of future CKM conditions and premature death.[76, 133, 134] Technology integration such as clinical decision support systems, wearable devices, and mobile applications can enhance communications and timely referrals between primary and specialist care teams, empower self-management, and facilitate shared decision-making.[76, 133, 135] However, there are ongoing concerns with digital care including data privacy and confidentiality, health literacy, accessibility, user-friendliness, and cost-effectiveness.[76, 135] Precision medicine approaches incorporating genomic and phenotyping information to guide management are being advocated, although examples of effective care models remain to be demonstrated.[136-138]

To reduce health inequality in access to evidence-based interventions, especially in resource-limited settings, standardization of evaluation and management procedures is needed. Figures 6 and 7 propose an inclusive management framework for people with obesity and CKD.[139] The framework outlines a comprehensive evaluation followed by an individualized management plan including person-centered goal setting and the optimal integration of dietary modification, physical activity, pharmacotherapy, and surgical interventions. Its implementation will depend on the presence of CKM risk factors, available resources, and shared treatment goals. In all settings, dietary modification and physical activity interventions remain as foundation therapy for CKD and are complementary to medical and surgical interventions for obesity.[91, 140] Adopting a weight-neutral approach that focuses on health outcomes and health behaviors may be useful to support people who do not wish to focus on weight *per se*. A structured care model that integrates evidence-based technologies and principles of precision medicine will enable person-centered decision-making and thus may offer a sustainable strategy for managing the complexity of obesity and CKD. To this end, public health policies that create a health-enabling ecosystem with capacity to facilitate the implementation of this integrated obesity and CKD management framework with health economic evaluation are eagerly awaited.

CONCLUSIONS AND FUTURE DIRECTIONS

Obesity and CKD are increasingly prevalent conditions that often coexist, presenting significant challenges for prevention and management. As highlighted throughout this conference, there is a growing awareness that CKD may be present in individuals living with obesity, yet many questions remain regarding early detection, optimal screening strategies, and the most effective ways to assess obesity-related kidney risk. Addressing these uncertainties is critical to improving long-term outcomes, particularly given the rising rates of obesity among children and young adults.

There is broad consensus that dietary changes, increased physical activity, and behavioral modifications form the foundation of obesity management in CKD. However, the most effective strategies for sustaining meaningful, long-term changes remain elusive. Pharmacotherapies such as GLP-1 RA offer substantial benefits, yet questions persist regarding their long-term safety, particularly for younger individuals and those with advanced CKD. Kidney transplant patients require careful consideration to avoid dehydration and to ensure the tolerability of these therapies alongside immunosuppressive regimens. While randomized controlled trials in this area are unlikely, expert opinion and retrospective data may provide valuable guidance moving forward.

A multidisciplinary approach—incorporating nephrology, endocrinology, hepatology, and dietetics—will be essential in refining treatment strategies and expanding the evidence base. Notably, limited data exist on how successful interventions for weight loss impact kidney function, particularly in advanced CKD. Future research should explore whether targeted interventions can not only slow progression but potentially reverse kidney dysfunction, addressing whether it is ever too late to intervene.

The urgency of preventing obesity early in life cannot be overstated. The long-term metabolic and kidney consequences of early-onset obesity are profound, emphasizing the need for preventive strategies and early interventions. At the same time, significant gaps remain in our understanding of global epidemiology, treatment efficacy, and healthcare implementation, particularly in underrepresented regions.

While many uncertainties remain, with continued collaboration, research, and innovation, strategies for managing obesity and CKD will continue to evolve and improve, with the aim of improving patient outcomes across the lifespan.

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Table 1. Key knowledge gaps and research priorities in obesity and CKD

	Knowledge Gaps and Key Questions	**Research Strategies**

Epidemiology and pathophysiology	• What are the trajectories of eGFR decline in patients with obesity and:
	  ○ preexisting kidney disease,
	  ○ kidney transplant,
	  ○ or premature birth or small-for-gestational-age
	• What is the optimal measure or biomarker for monitoring the effect of obesity on kidney function?
	• In individuals with obesity what is the best estimation of kidney function, comparing to the gold standard of measured GFR?
	• What is the role of inflammation in the pathophysiology of obesity and CKD?
	• In individuals with obesity, how do environmental stressors—including air pollution, socioeconomic stressors (including food deserts), microplastics, heat stress, and the consumption of ultra-processed food—influence the risk of CKD?
		• Examine the role of obesity in altering the rate of eGFR decline in patients with kidney disease.
		• Evaluate the role of obesity on graft function and cardiovascular outcomes in kidney transplant recipients.
		• Measure metabolic changes and energy consumption in relation to obesity in patients with kidney damage.
		• Identify the age range during childhood when BMI is most predictive of adverse outcome and discern whether trajectory in children or young adults is more important than single time point of BMI.

Evaluation of kidney function and adiposity	• What are the impacts of muscle mass and muscle quality on kidney disease progression in individuals with obesity and CKD? • Would addition of kidney parameters improve the utility of King's College Criteria, or is a new scoring system needed to determine treatment needs? • Can assessment of inflammatory markers or visceral organ adiposity aid in predicting outcomes?	• Compare the predictive ability of different anthropometric measures (Table 2) and body composition tools (alone or in combination) in those with obesity and CKD. • Use different modalities to investigate muscle changes with weight loss. • Incorporate metrics beyond BMI and WC to assess the efficacy of different interventions especially in those with sarcopenic and visceral adiposity (include patient perspectives). • Identify alternative options to measuring albuminuria (UACR vs 24-hour urine collection) in those undergoing weight loss (include patient feedback).
Non-medical interventions	• How do we implement and monitor non-medical interventions for people with CKD and obesity? • Do personalized interventions based on individual risk assessment improve outcomes in CKD? • Which PROMs are most relevant and important?	• Conduct qualitative surveys to understand the challenges that people with obesity and CKD face in implementing non-medical interventions. • Develop an obesity and CKD core outcome set including patient-reported outcomes and tools for measurement. • Implement diet and exercise interventions and monitoring plans for people with CKD and obesity based on CKM risks and phenotypes in real-world practice. • Conduct pragmatic implementation trials (population: individuals with CKD and obesity; intervention: personalized based on assessment and

		patient goals; comparison: usual care; outcomes: changes in eGFR slope, albuminuria, health-related quality of life, physical function).
Medical interventions	• In patients with CKD, what are the relative efficacies of surgical and newer medical therapies on weight loss, kidney and cardiometabolic outcomes, and safety? • In children, can drug therapy obviate surgery or delay its need until maturity? • What are the efficacy and safety profiles of obesity interventions in groups with need but where data are lacking (extremes of age, diverse demographics, advanced CKD [including dialysis and kidney transplant])?	• In studying weight loss interventions studies, routinely include people with CKD and outcomes of kidney function. • Conduct head-to-head comparisons of the efficacy and safety of drug therapies before metabolic and bariatric surgery. • Evaluate the efficacy and safety of drug therapies after metabolic and bariatric surgery. • Initiate implementation research on the best approaches to ensure those in need are offered effective therapies with minimal side effects, non-persistence is minimized, and drug dosage is appropriately tailored. • Capture real-world epidemiologic and health economic data to analyze the full range benefits and costs of weight loss interventions in people living with obesity and CKD.
Care models	• How much weight loss is needed to improve health outcomes in people with obesity and CKD? • How to scale up the assessment and monitoring for PROMs, PREMs, physical function, and functionality in routine practice?	• Employ RCTs, observational studies, and evidence reviews to define relationships between percentage weight loss on clinical outcomes and PROMs in people with obesity and CKD with a duration of at least 12 months and a long follow-up.

	• How to enhance the implementation of non-medical interventions in routine practice, from the perspectives of sustainability and health economics?	• Conduct RCTs to delineate the percentage of weight loss target for CKD outcomes (UACR, eGFR) and health-related quality of life. • Evaluate the effectiveness and sustainability of obesity care interventions that include a combination of behavioral, pharmacological, and surgical interventions, together with dietary and physical activity interventions, in people with obesity and CKD. • Evaluate self-management strategies and behavioral modifications for sustaining health outcomes in people with obesity and CKD. • Evaluate mobile assistive-technology tools for sustainable weight reduction in people with obesity and CKD, including in large-scale implementation models.

Abbreviations: ACR, albumin to creatinine ratio; BMI, body mass index; CKD, chronic kidney disease; eGFR, estimated glomerular filtration rate; PREMs, patient-reported experience measures; PROMs, patient-reported outcome measures; RCTs, randomized clinical trials; WC, waist circumference; UACR, urine albumin-to-creatinine ratio.

Table 2. The comparison of multiple anthropometric and imaging modalities for assessing obesity

Metric	Remarks	Strengths	Limitations
Body mass index (BMI)	Measures weight relative to height; used as a general obesity indicator	Simple, widely used, correlates with body fat in population studies	Does not reflect fat distribution or differentiate between fat and muscle mass; affected by high muscle mass
Waist circumference (WC)	Measures abdominal fat	Easy to measure, correlates with visceral fat, good indicator of abdominal obesity	Unable to distinguish between subcutaneous and visceral adipose tissue; not adjusted for height or body structure
Waist-to-height ratio	Ratio of waist circumference to height	Better predictor of cardiovascular risk than BMI; accounts for body frame	Lacks cut-off points for different populations; does not differentiate between subcutaneous and visceral fat
Waist-to-hip ratio	Ratio of waist circumference to hip circumference	Correlates with visceral fat and metabolic risk factors; useful in assessing cardiovascular risk	Limited by measurement accuracy; less useful in older adults due to hip shape changes
Percentage of body fat	Represents total body fat as a percentage of body weight	More accurate measure of total body fat than BMI	Requires specialized equipment (calipers, bioelectrical impedance, DEXA); may vary based on hydration and other factors
A body shape index	Incorporates waist circumference and BMI	Predicts mortality risk independently of BMI; accounts for central obesity	Complex calculation that is less intuitive than BMI or WC; limited use in clinical practice
Body roundness index	Considers body shape to estimate fat distribution	Provides detailed body shape analysis; can be used to estimate body fat and health risk	Requires specific calculations and validation in different populations; limited research
Relative fat mass	Uses height and waist circumference to estimate body fat	Simple and correlates well with body fat; validated in various populations	Lacks comprehensive data across diverse populations; may not be as accurate for athletes or individuals with varying body composition
Visceral adiposity index	Estimation of visceral fat based on BMI, triglycerides, and HDL	Correlates well with visceral fat and cardiometabolic risk	Requires lab results (triglycerides, HDL) that are not easily calculated in routine practice

Weight-adjusted waist index	Adjusts waist circumference for weight	Helps differentiate between weight and abdominal obesity	Not well-established in clinical guidelines; lacks widespread research
Lipid accumulation product	Based on waist circumference and fasting triglycerides	Reflects visceral fat accumulation; associated with cardiometabolic risk	Requires blood tests; may not be as useful in populations with low prevalence of metabolic syndrome

BMI, body mass index; DEXA, dual x-ray absorptiometry; HDL high-density lipoprotein; WC, waist circumference

Table 3. Effective nutrition-based interventions for people with obesity and CKD

Nutrition intervention	Physical activity intervention	CKD category	Other support provided	Suggested adaptation for CKD G4–G5
Mediterranean diet with energy reduction				
Encourages whole foods predominantly from plant sources including grains, fruits, vegetables, pulses, nuts, and seeds Small servings of poultry, fish, or seafood 4–5 times a week; red meat once a week or fewer (fish or poultry to replace beef or lamb) Daily olive oil (in salad dressings and sauces); handful of nuts 3 times a week to once a day Reduce processed foods, sweets, pastries, and cakes **Energy intake** Baseline energy intake minus 500–600 kcal/day or total energy intake 1500 kcal/day for women and 1800 kcal/day for men **Macronutrients and Fiber** 40–45% carbohydrates (low glycemic index choices) 30–35 g fiber/day 20% protein (plant sources preferred) 35–40% fat (plant sources preferred)	Simple advice to increase activity to 150 minutes per week (Orazio et al., 2011)[81]	Post kidney transplantation	Written information Pictorial guide	For G4 and G5: modify protein intake to 0.8 g protein/kg body weight/day Likely safe if supervised by a clinical team; may require closer monitoring of potassium initially Increase protein intake in CKD G5D from plant sources (pulses and nuts) and poultry, fish, and seafood to meet protein goal of 1–1.2 g/kg body weight/day; maintain fiber intake at 30–35 g/day
	Intensive support for physical activity and behavior change (Díaz-López et al., 2021)[80]	Included non-CKD and CKD G1–G3b	List of lower-energy foods Mediterranean diet food group servings Used adapted Mediterranean diet checklist	
	None (Tirosh et al., 2013)[82]	CKD G3 (31% of study participants had eGFR 30–60 ml/min per 1.73 m^2)	Dietitian led group sessions fortnightly for 2 months and then every 6 weeks up to 18 months Telephone support Education for spouses	
Low-fat diet with energy reduction				

Consume grains, vegetables, fruits, and legumes and limit intake of additional fats, sweets, and high-fat snacks **Energy intake** Baseline energy intake minus 500–600 kcal/day or total energy intake 1500 kcal/day for women and 1800 kcal/day for men **Macronutrients** Fat 25–30% Protein 1.0–1.2g/kg adjusted body weight used in CKD G1–G3b and G5D, or 0.6–0.8g/kg adjusted body weight in CKD G4	None (Tirosh et al., 2013)[82]	CKD G3 (31% of study participants had eGFR 30–60 ml/min per 1.73m^2)	Dietitian led group sessions fortnightly for 2 months and then every 6 weeks up to 18 months Telephone support Education for spouses	Modify protein intake in CKD G5D, often as 1–2x 100 g servings of lean poultry, tuna, or 2 eggs
	Moderate-intensity aerobic exercises 3 to 5 days/week for 30 minutes (Kittiskulnam et al., 2014)[83]	CKD G1–G4 with proteinuria	Nutritionist and clinician delivered intervention	
	None (Morales et al., 2003)[84]	CKD ~G1–G3a with proteinuria	No additional information provided	
	Home-based exercise 5 days a week Start at 50 mins/week and increase to 175 mins/week (Look Ahead Research Group, 2014)[77]	Non-CKD and CKD G1–G2 with type 2 diabetes (excluded CKD G3–G5)	Included 1–2 meal replacement products per day (energy 1200–1800 kcal/day) Intensive support provided for 18 months Refresher group program and/or orlistat offered if target weight loss not met Other supports included exercise or cooking classes	
Healthy diet with energy reduction				
Energy intake No specific nutrient intake prescribed; 10–15% reduction in total daily energy intake from baseline **Dietary pattern** Large amounts of fresh, unprocessed plant foods along with moderate levels of lean protein and fats such as omega-3 and monounsaturated fats	In exercise arms: supervised aerobic exercise 30–45 minutes 3x per week (Ikizler et al., 2018)[85]	CKD G3–G4	2x2 design, exercise or energy reduction or both or control Dietitian-led dietary intervention: avoid high-calorie meals rich in processed, easily digestible, quickly absorbable foods and drinks as a part of a CKD diet.	None for CKD G4 Monitor for hypotension with weight loss. For CKD G5D additional monitoring for electrolytes and dry weight adjustment. Additional protein may be required, often as 1–2 x 100 g

				servings of lean poultry, tuna, or 2 eggs
Low-energy diets using formula meal replacements				
Energy intake 800–900 kcal/day and 75–80 g protein **Composition** 3–4 commercially produced standardized and micronutrient-fortified meal replacement shakes, bars, and soups per day	Remotely delivered exercise support with graded exercise program after dietary intervention ceased (Conley et al., 2025)[86]	CKD G1–G3b	Plus 1 piece fruit and 1 low-energy small meal and 2 cups of non-starchy vegetables per day Fortnightly dietitian support	Safe with clinical supervision Feasible over short term (up to 3 months) For CKD G5D additional close monitoring for electrolytes and dry weight adjustment. Additional protein may be required, often as 1–2 x 100 g servings of lean poultry, fish, or 2 eggs
	None (Friedman et al., 2013)[87]	CKD G3b–G4[87]	Plus one lean meal, 50g carbohydrate per day	
Low carbohydrate diet without energy reduction				
Energy intake Total energy was not limited **Macronutrient intake** 20 g carbohydrate per day for 2 months, then gradually increase to max 120 g carbohydrate per day No limits on fat and protein Choose vegetarian sources of fat and protein; avoid trans fats	None (Tirosh et al., 2013)[82]	CKD G3 (31% of study participants had eGFR 30–60 ml/min per 1.73 m^2)	Dietitian-led group sessions fortnightly for 2 months and then every 6 weeks up to 18 months Telephone support Education for spouses	Not suitable for CKD G4–G5
New Nordic renal diet without energy reduction				
Whole food approach: 80% plant-based, 20% animal products 1–2 vegetarian days/week Fish 2 or more servings/week Poultry 2 servings/week No red meat 300 g fruit per day, 20g nuts per day Protein 0.8 g/kg body weight/day	None (Misella Hansen et al., 2023) [88]	CKD G3b–G4, eGFR 20–45 ml/min per $1.73m^2$ [88]	Choose mostly whole grains, legumes, and root vegetables as well as in-season vegetables Food boxes and recipes were delivered to households each week Dietitian home visit was offered	Increase protein intake in CKD G5D from plant sources (pulses and nuts) and poultry, fish, and seafood to meet protein goal of 1–1.2g/kg body weight/day Maintain fiber intake at 30–35 g/day

Abbreviations: CKD, chronic kidney disease; eGFR, estimated glomerular filtration rate

Table 4. Key randomized clinical trial data on kidney outcomes from phase 3 or 4 trials of GLP-1 RA

Author	**Description**	**Main kidney outcome definition**	**HR (95% CI) associated with GLP-1 RA versus comparator**
Sattar et al., 2021[92]	Meta-analysis of kidney outcomes with GLP-1 RA in 8 trials in type 2 diabetes[a]	Severely increased albuminuria, doubling of serum creatinine, or at least 40% decline in eGFR, kidney replacement therapy, or death due to kidney disease	0.79 (0.73, 0.87), *P* < 0.0001
Perkovic et al., 2024[93]	FLOW: semaglutide 1 mg weekly versus placebo in type 2 diabetes with CKD	Kidney failure (dialysis, transplantation, or eGFR <15 ml/min per 1.73m^2), ≥50% eGFR reduction from baseline, or kidney or cardiovascular death	0.76 (0.66 to 0.88), *P* = 0.0003 Excluding CVD death from endpoint HR: 0.79; 95% CI: 0.66–0.94
Colhoun et al., 2024[94]	SELECT: semaglutide 2.4 mg weekly vs placebo in people with overweight or obesity and prior CVD but without type 2 diabetes	Death from kidney causes, initiation of chronic kidney replacement therapy (dialysis or transplantation), onset of persistent eGFR <15 ml/min per 1.73 m^2, persistent ≥50% reduction in eGFR compared with baseline or onset of persistent severely increased albumunuria	0.78 (0.63–0.96), *P* = 0.02
Badve et al., 2025[95]	Meta-analysis of all GLP-1 RA	Kidney failure (i.e., kidney replacement therapy or a persistent estimated glomerular	0.81 (0.72–0.92)

	trials including SELECT and FLOW[b]	filtration rate [eGFR] <15 mL/min per 1.73 m^2), a sustained reduction in eGFR from baseline by 50% or more, or death due to kidney disease	$P < 0.001$

CVD, cardiovascular disease; eGFR, estimated glomerular filtration rate; GLP-1 RA, glucagon-like peptide-1 receptor agonists; FLOW, Evaluate Renal Function with Semaglutide Once Weekly; SELECT, Semaglutide Effects on Cardiovascular Outcomes in People With Overweight or Obesity.

[a]Combined data from ELIXA (lixisenatide),[141] LEADER (liraglutide),[142, 143] SUSTAIN-6 (semaglutide),[144] EXSCEL (exenatide),[145] REWIND (dulaglutide),[146, 147] Harmony Outcomes (albiglutide),[148] PIONEER 6 (semaglutide),[149] and AMPLITUDE-O (efpeglenatide) trials[150]

[b]Combined data from ELIXA (lixisenatide),[141] LEADER (liraglutide),[142, 143] SUSTAIN-6 (semaglutide),[144] EXSCEL (exenatide),[145] REWIND (dulaglutide),[146, 147] Harmony Outcomes (albiglutide),[148] AMPLITUDE-O (efpeglenatide),[150] FLOW (semaglutide),[93] and SELECT (semaglutide)[94] trials

Figure Legends

Figure 1. Metabolic domino as a conceptual model for the flow of events and chain reactions associated with obesity and chronic kidney disease. The order of risk factors and outcomes is schematic and will differ between individuals based on genetic predispositions and lived environments. Changes in diet and physical activity level can lead to obesity and insulin resistance, followed by hyperglycemia, hypertension, and dyslipidemia. Worsening kidney function as evidenced by albuminuria and reduced glomerular filtration rate can lead to chronic disease and its sequelae. Progression of the atherosclerotic process can lead to cardiovascular events such as ischemic heart diseases or cerebrovascular disorders. Preclinical and clinical data indicate that treatments that inhibit the renin angiotensin system, such as angiotensin receptor blockers, can suppress the onset of type 2 diabetes and, when administered even earlier in the metabolic domino, reduce the development of hypertension in at-risk individuals. CKD, chronic kidney disease; CVD, cardiovascular disease; MAFLD, metabolic dysfunction–associated fatty liver disease. © Jennifer N. Gentry.

Figure 2. Hypothesized mechanisms for hyperglycemia-induced hypoxia leading to kidney tissue damage. Hyperglycemia following insulin resistance may induce single-nephron hyperfiltration, tubular growth, and upregulation of sodium-glucose transporters. This may lead to an increase in sodium reabsorption, increasing adenosine triphosphate (ATP) demand and oxygen (O_2) consumption. Microvascular damage caused by hyperglycemia could decrease renal perfusion and O_2 delivery and shift glucose oxidation toward the less efficient oxidation of free fatty acids, resulting in decreased generation of ATP per O_2 molecule. AMPK, AMP-activated

protein kinase; ATP, adenosine triphosphate; ECM, extracellular matrix; iR, insulin resistance; RAAS, renin–angiotensin–aldosterone system; ROS, reactive oxygen species; SGLT, sodium-glucose cotransporter. Modified from Hesp et al., 2020.[35]

Figure 3. Comparison of cystatin C and creatinine for kidney function assessment. GFR, glomerular filtration rate.

Figure 4. Comparison of pharmacotherapy with metabolic and bariatric surgery for weight loss in chronic kidney disease. Factors such as patient life goals, comorbidities, and ability to engage in physical activity will inform the therapeutic strategy. Metabolic and bariatric surgery may not be appropriate for all ages.

Figure 5. Person-Centered Care in action: A collaborative care network for CKD and obesity. The inner circle is risk factors, comorbidities, and outcomes; the middle circle is the team (kidney-specific professional expertise is preferred); the outer circle refers to broader determinants of health and resources. AI, artificial intelligence; Apps, mobile health applications; CKM, cardiovascular–kidney–metabolic; PCP, primary care practitioners; PREMs, patient-reported experience measures; PROMs, patient-reported outcome measures.

Figure 6. Care delivery in people with CKD and obesity: A framework to guide decision-making. Modified from Vallis et al., 2013.[139]

Figure 7. Care model in people with CKD and obesity including non-medical, medical, and surgical interventions and the relationship between treatment options. All interventions include nutrition, physical activity, and behavior change, with additional psychological, surgical, and pharmacological treatment considered based on comprehensive assessment. Consider digital or virtual technologies and centralized expertise for training and supporting local teams to provide expert involvement in assessment, decision-making, and follow-up. BMI, body mass index; MAFLD, metabolic dysfunction–associated fatty liver disease; PREMs, patient-reported experience measures; PROMs, patient-reported outcome measures; MAFLD, metabolic dysfunction–associated fatty liver disease; UACR, urine albumin-to-creatinine ratio; WC, waist circumference; WHR, waist-to-hip ratio; WHtR, waist-to-height ratio. © Jennifer N. Gentry.

REFERENCES

1. World Obesity Federation. World Obesity Atlas 2024. Available at: https://www.worldobesity.org/resources/resource-library/world-obesity-atlas-2024. Accessed July 11, 2024.

2. Kovesdy CP. Epidemiology of chronic kidney disease: an update 2022. *Kidney Int Suppl* 2022; **12:** 7-11.

3. Global, regional, and national burden of chronic kidney disease, 1990-2017: A systematic analysis for the Global Burden of Disease Study 2017. *Lancet* 2020; **395:** 709-733.

4. Câmara NO, Iseki K, Kramer H, *et al.* Kidney disease and obesity: Epidemiology, mechanisms and treatment. *Nat Rev Nephrol* 2017; **13:** 181-190.

5. Stenvinkel P, Zoccali C, Ikizler TA. Obesity in CKD--what should nephrologists know? *J Am Soc Nephrol* 2013; **24:** 1727-1736.

6. Lee-Boey J-WS, Tan J-K, Lim Z-F, *et al.* Obesity-related glomerulopathy: how it happens and future perspectives. *Diabetic Medicine* 2025; **42:** e70042.

7. Conley MM, McFarlane CM, Johnson DW, *et al.* Interventions for weight loss in people with chronic kidney disease who are overweight or obese. *Cochrane Database Syst Rev* 2021; **3:** Cd013119.

8. Chintam K, Chang AR. Strategies to treat obesity in patients with CKD. *Am J Kidney Dis* 2021; **77:** 427-439.

9. Mingrone G, Panunzi S, De Gaetano A, *et al.* Metabolic surgery versus conventional medical therapy in patients with type 2 diabetes: 10-year follow-up of an open-label, single-centre, randomised controlled trial. *Lancet* 2021; **397:** 293-304.

10. Davies M, Færch L, Jeppesen OK, *et al.* Semaglutide 2·4 mg once a week in adults with overweight or obesity, and type 2 diabetes (STEP 2): A randomised, double-blind, double-dummy, placebo-controlled, phase 3 trial. *Lancet* 2021; **397:** 971-984.

11. Jastreboff AM, Aronne LJ, Ahmad NN, *et al.* Tirzepatide once weekly for the treatment of obesity. *N Engl J Med* 2022; **387:** 205-216.

12. von Scholten BJ, Persson F, Svane MS, *et al.* Effect of large weight reductions on measured and estimated kidney function. *BMC Nephrology* 2017; **18:** 52.

13. Chang AR, George J, Levey AS, *et al.* Performance of glomerular filtration rate estimating equations before and after bariatric surgery. *Kidney Med* 2020; **2:** 699-706.e691.

14. World Health Organization. Obesity and Overweight. Available at: https://www.who.int/news-room/fact-sheets/detail/obesity-and-overweight. Accessed November 19, 2024.

15. World Health Organization. Knowledge Action Portal on NCDs. Available at: https://knowledge-action-portal.com/en/content/obesity-and-overweight. Accessed February 20, 2025.

16. Boutari C, Mantzoros CS. A 2022 update on the epidemiology of obesity and a call to action: As its twin COVID-19 pandemic appears to be receding, the obesity and dysmetabolism pandemic continues to rage on. *Metabolism* 2022; **133:** 155217.

17. NCD Risk Factor Collaboration (NCD-RisC). Worldwide trends in underweight and obesity from 1990 to 2022: A pooled analysis of 3663 population-representative studies with 222 million children, adolescents, and adults. *Lancet* 2024; **403:** 1027-1050.

18. Wang Y, Li F, Chu C, *et al.* Early life body mass index trajectories and albuminuria in midlife: A 30-year prospective cohort study. *EClinicalMedicine* 2022; **48:** 101420.

19. Al Saikhan L, Chaturvedi N, Ghosh AK, *et al.* Adulthood adiposity affects cardiac structure and function in later life. *Eur Heart J* 2024; **45:** 3060-3068.

20. Hsu CY, McCulloch CE, Iribarren C, *et al.* Body mass index and risk for end-stage renal disease. *Ann Intern Med* 2006; **144:** 21-28.

21. Chang AR, Grams ME, Ballew SH, *et al.* Adiposity and risk of decline in glomerular filtration rate: meta-analysis of individual participant data in a global consortium. *BMJ* 2019; **364:** k5301.

22. Fox CS, Larson MG, Leip EP, *et al.* Predictors of new-onset kidney disease in a community-based population. *Jama* 2004; **291:** 844-850.

23. Wang Y, Chen X, Song Y, *et al.* Association between obesity and kidney disease: A systematic review and meta-analysis. *Kidney Int* 2008; **73:** 19-33.

24. Zhang X, Liu J, Ni Y, *et al.* Global prevalence of overweight and obesity in children and adolescents: A systematic review and meta-analysis. *JAMA Pediatr* 2024; **178:** 800-813.

25. Afshin A, Forouzanfar MH, Reitsma MB, *et al.* Health effects of overweight and obesity in 195 countries over 25 years. *N Engl J Med* 2017; **377:** 13-27.

26. Cleland V, Tian J, Buscot MJ, *et al.* Body-mass index trajectories from childhood to mid-adulthood and their sociodemographic predictors: Evidence from the International Childhood Cardiovascular Cohort (i3C) Consortium. *EClinicalMedicine* 2022; **48:** 101440.

27. Silverwood RJ, Pierce M, Thomas C, *et al.* Association between younger age when first overweight and increased risk for CKD. *J Am Soc Nephrol* 2013; **24:** 813-821.

28. Kramer HJ, Saranathan A, Luke A, *et al.* Increasing body mass index and obesity in the incident ESRD population. *J Am Soc Nephrol* 2006; **17:** 1453-1459.

29. Lu JL, Kalantar-Zadeh K, Ma JZ, *et al.* Association of body mass index with outcomes in patients with CKD. *J Am Soc Nephrol* 2014; **25:** 2088-2096.

30. Reynolds K, Gu D, Muntner P, *et al.* Body mass index and risk of ESRD in China. *Am J Kidney Dis* 2007; **50:** 754-764.

31. Stenvinkel P, Shiels PG, Kotanko P, *et al.* Harnessing evolution and biomimetics to enhance planetary health: kidney insights. *J Am Soc Nephrol* 2024; **36:** 311-321.

32. Fruhbeck G, Kiortsis DN, Catalan V. Precision medicine: diagnosis and management of obesity. *Lancet Diabetes Endocrinol* 2018; **6:** 164-166.

33. de Oliveira RB, Pelepenko LE, Masaro DA, *et al.* Effects of microplastics on the kidneys: a narrative review. *Kidney Int* 2024; **106:** 400-407.

34. Zhu P, Herrington WG, Haynes R, *et al.* Conventional and genetic evidence on the association between adiposity and CKD. *J Am Soc Nephrol* 2021; **32:** 127-137.

35. Hesp AC, Schaub JA, Prasad PV, *et al.* The role of renal hypoxia in the pathogenesis of diabetic kidney disease: A promising target for newer renoprotective agents including SGLT2 inhibitors? *Kidney Int* 2020; **98:** 579-589.

36. Yıldız AB, Vehbi S, Copur S, *et al.* Kidney and liver fat accumulation: from imaging to clinical consequences. *J Nephrol* 2024; **37:** 483-490.

37. Kanbay M, Copur S, Demiray A, *et al.* Fatty kidney: a possible future for chronic kidney disease research. *Eur J Clin Invest* 2022; **52:** e13748.

38. Chertow GM, Chang AM, Felker GM, *et al.* IL-6 inhibition with clazakizumab in patients receiving maintenance dialysis: A randomized phase 2b trial. *Nat Med* 2024; **30:** 2328-2336.

39. Ridker PM. From RESCUE to ZEUS: Will interleukin-6 inhibition with ziltivekimab prove effective for cardiovascular event reduction? *Cardiovasc Res* 2021; **117:** e138-e140.

40. Stenvinkel P, Gillespie IA, Tunks J, *et al.* Inflammation modifies the paradoxical association between body mass index and mortality in hemodialysis patients. *J Am Soc Nephrol* 2016; **27:** 1479-1486.

41. Rubino F, Cummings DE, Eckel RH, *et al.* Definition and diagnostic criteria of clinical obesity. *Lancet Diabetes Endocrinol* 2025; **13:** 221-262.

42. Madero M, Katz R, Murphy R, *et al.* Comparison between different measures of body fat with kidney function decline and incident CKD. *Clin J Am Soc Nephrol* 2017; **12:** 893-903.

43. Pinto-Sietsma SJ, Navis G, Janssen WM, *et al.* A central body fat distribution is related to renal function impairment, even in lean subjects. *Am J Kidney Dis* 2003; **41:** 733-741.

44. Stefansson VTN, Schei J, Solbu MD, *et al.* Metabolic syndrome but not obesity measures are risk factors for accelerated age-related glomerular filtration rate decline in the general population. *Kidney Int* 2018; **93:** 1183-1190.

45. Ozbek L, Abdel-Rahman SM, Unlu S, *et al.* Exploring Adiposity and Chronic Kidney Disease: Clinical Implications, Management Strategies, Prognostic Considerations. *Medicina* 2024; **60:** 1668.

46. Navaneethan SD, Schold JD, Kirwan JP, *et al.* Metabolic syndrome, ESRD, and death in CKD. *Clin J Am Soc Nephrol* 2013; **8:** 945-952.

47. Kramer H, Shoham D, McClure LA, *et al.* Association of waist circumference and body mass index with all-cause mortality in CKD: The REGARDS (Reasons for Geographic and Racial Differences in Stroke) Study. *Am J Kidney Dis* 2011; **58:** 177-185.

48. Marcelli D, Brand K, Ponce P, *et al.* Longitudinal changes in body composition in patients after initiation of hemodialysis therapy: Results from an international cohort. *J Ren Nutr* 2016; **26:** 72-80.

49. Caetano C, Valente A, Oliveira T, *et al.* Body composition and mortality predictors in hemodialysis patients. *J Ren Nutr* 2016; **26:** 81-86.

50. Kanbay M, Copur S, Siriopol D, *et al.* The risk for chronic kidney disease in metabolically healthy obese patients: A systematic review and meta-analysis. *Eur J Clin Invest* 2023; **53:** e13878.

51. Kataoka H, Nitta K, Hoshino J. Visceral fat and attribute-based medicine in chronic kidney disease. *Front Endocrinol (Lausanne)* 2023; **14:** 1097596.

52. Bamba R, Okamura T, Hashimoto Y, *et al.* The Visceral Adiposity Index Is a Predictor of Incident Chronic Kidney Disease: A Population-Based Longitudinal Study. *Kidney Blood Press Res* 2020; **45:** 407-418.

53. Bellafronte NT, de Queirós Mattoso Ono A, Chiarello PG. Sarcopenic obesity in chronic kidney disease: challenges in diagnosis using different diagnostic criteria. *Med Princ Pract* 2021; **30:** 477-486.

54. Noori N, Hosseinpanah F, Nasiri AA, *et al.* Comparison of overall obesity and abdominal adiposity in predicting chronic kidney disease incidence among adults. *J Ren Nutr* 2009; **19:** 228-237.

55. Wei S, Nguyen TT, Zhang Y, *et al.* Sarcopenic obesity: epidemiology, pathophysiology, cardiovascular disease, mortality, and management. *Frontiers in Endocrinology* 2023; **14:** 1185221.

56. Duarte MP, Almeida LS, Neri SG, *et al.* Prevalence of sarcopenia in patients with chronic kidney disease: a global systematic review and meta-analysis. *Journal of cachexia, sarcopenia and muscle* 2024; **15:** 501-512.

57. Seo DH, Suh YJ, Cho Y, *et al.* Effect of low skeletal muscle mass and sarcopenic obesity on chronic kidney disease in patients with type 2 diabetes. *Obesity* 2022; **30:** 2034-2043.

58. Schwartz P, Capotondo MM, Quaintenne M, *et al.* Obesity and glomerular filtration rate. *Int Urol Nephrol* 2024; **56:** 1663-1668.

59. Navaneethan SD, Malin SK, Arrigain S, *et al.* Bariatric surgery, kidney function, insulin resistance, and adipokines in patients with decreased GFR: a cohort study. *Am J Kidney Dis* 2015; **65:** 345-347.

60. Lemoine S, Guebre-Egziabher F, Sens F, *et al.* Accuracy of GFR estimation in obese patients. *Clin J Am Soc Nephrol* 2014; **9:** 720-727.

61. Delanaye P, Radermecker RP, Rorive M, *et al.* Indexing glomerular filtration rate for body surface area in obese patients is misleading: concept and example. *Nephrol Dial Transplant* 2005; **20:** 2024-2028.

62. Fotheringham J, Weatherley N, Kawar B, *et al.* The body composition and excretory burden of lean, obese, and severely obese individuals has implications for the assessment of chronic kidney disease. *Kidney Int* 2014; **86:** 1221-1228.

63. Russel WA, Fu EL, Bosi A, *et al.* Obesity, underweight, and accuracy of eGFR using cystatin C and creatinine in a Northern European population. *J Am Soc Nephrol* 2025.

64. Chagnac A, Friedman AN. Measuring Albuminuria in Individuals With Obesity: Pitfalls of the Urinary Albumin-Creatinine Ratio. *Kidney Med* 2024; **6:** 100804.

65. Lee JH, McDonald EO, Harhay MN. Obesity management in kidney transplant candidates: current paradigms and gaps in knowledge. *Advances in chronic kidney disease* 2021; **28:** 528-541.

66. Oniscu GC, Abramowicz D, Bolignano D, *et al.* Management of obesity in kidney transplant candidates and recipients: A clinical practice guideline by the DESCARTES Working Group of ERA. *Nephrol Dial Transplant* 2021; **37:** i1-i15.

67. Krishnan N, Higgins R, Short A, *et al.* Kidney Transplantation Significantly Improves Patient and Graft Survival Irrespective of BMI: A Cohort Study. *Am J Transplant* 2015; **15:** 2378-2386.

68. Segev DL, Simpkins CE, Thompson RE, *et al.* Obesity impacts access to kidney transplantation. *J Am Soc Nephrol* 2008; **19:** 349-355.

69. Lavenburg LU, Kim Y, Weinhandl ED, *et al.* Trends, social context, and transplant implications of obesity among incident dialysis patients in the United States. *Transplantation* 2022; **106:** e488-e498.

70. Chang JH, Mushailov V, Mohan S. Obesity and kidney transplantation. *Curr Opin Organ Transplant* 2023; **28:** 149-155.

71. Harhay MN, Ranganna K, Boyle SM, *et al.* Association Between Weight Loss Before Deceased Donor Kidney Transplantation and Posttransplantation Outcomes. *Am J Kidney Dis* 2019; **74:** 361-372.

72. Harhay MN, Chen X, Chu NM, *et al.* Pre-kidney transplant unintentional weight loss leads to worse post-kidney transplant outcomes. *Nephrol Dial Transplant* 2021; **36:** 1927-1936.

73. Kostro JZ, Bzoma B, Proczko-Stepaniak M, *et al.* Kidney transplantation in patients after bariatric surgery: high-volume bariatric and transplant center experience. *Transplant Proc* 2022; **54:** 955-959.

74. Yemini R, Nesher E, Winkler J, *et al.* Bariatric surgery in solid organ transplant patients: Long-term follow-up results of outcome, safety, and effect on immunosuppression. *Am J Transplant* 2018; **18:** 2772-2780.

75. Bolignano D, Zoccali C. Effects of weight loss on renal function in obese CKD patients: a systematic review. *Nephrol Dial Transplant* 2013; **28 Suppl 4:** iv82-98.

76. KDIGO 2024 Clinical Practice Guideline for the Evaluation and Management of Chronic Kidney Disease. *Kidney Int* 2024; **105:** S117-s314.

77. The Look Ahead Research Group. Effect of a long-term behavioural weight loss intervention on nephropathy in overweight or obese adults with type 2 diabetes: a secondary analysis of the Look AHEAD randomised clinical trial. *Lancet Diabetes Endocrinol* 2014; **2:** 801-809.

78. Podadera-Herreros A, Arenas-de Larriva AP, Gutierrez-Mariscal FM, *et al.* Correction: Mediterranean diet as a strategy for preserving kidney function in patients with coronary heart disease with type 2 diabetes and obesity: a secondary analysis of CORDIOPREV randomized controlled trial. *Nutr Diabetes* 2024; **14:** 44.

79. Podadera-Herreros A, Arenas-de Larriva AP, Gutierrez-Mariscal FM, *et al.* Mediterranean diet as a strategy for preserving kidney function in patients with coronary heart disease with type 2 diabetes and obesity: a secondary analysis of CORDIOPREV randomized controlled trial. *Nutr Diabetes* 2024; **14:** 27.

80. Díaz-López A, Becerra-Tomás N, Ruiz V, *et al.* Effect of an intensive weight-loss lifestyle intervention on kidney function: a randomized controlled trial. *Am J Nephrol* 2021; **52:** 45-58.

81. Orazio LK, Isbel NM, Armstrong KA, *et al.* Evaluation of dietetic advice for modification of cardiovascular disease risk factors in renal transplant recipients. *J Ren Nutr* 2011; **21:** 462-471.

82. Tirosh A, Golan R, Harman-Boehm I, *et al.* Renal function following three distinct weight loss dietary strategies during 2 years of a randomized controlled trial. *Diabetes Care* 2013; **36:** 2225-2232.

83. Kittiskulnam P, Kanjanabuch T, Tangmanjitjaroen K, *et al.* The beneficial effects of weight reduction in overweight patients with chronic proteinuric immunoglobulin a nephropathy: a randomized controlled trial. *J Ren Nutr* 2014; **24:** 200-207.

84. Morales E, Valero MA, Leon M, *et al.* Beneficial effects of weight loss in overweight patients with chronic proteinuric nephropathies. *Am J Kidney Dis* 2003; **41:** 319-327.

85. Ikizler TA, Robinson-Cohen C, Ellis C, *et al.* Metabolic effects of diet and exercise in patients with moderate to severe CKD: a randomized clinical trial. *J Am Soc Nephrol* 2018; **29:** 250-259.

86. Conley MM, Mayr HL, Hepburn KS, *et al.* Low energy diets for obesity and CKD (SLOW-CKD Randomized Feasibility Study). *Kidney Int Rep* 2025.

87. Friedman AN, Chambers M, Kamendulis LM, *et al.* Short-term changes after a weight reduction intervention in advanced diabetic nephropathy. *Clin J Am Soc Nephrol* 2013; **8:** 1892-1898.

88. Misella Hansen N, Kamper AL, Rix M, *et al.* Health effects of the New Nordic Renal Diet in patients with stage 3 and 4 chronic kidney disease, compared with habitual diet: a randomized trial. *Am J Clin Nutr* 2023; **118:** 1042-1054.

89. MacLaughlin HL, Friedman AN, Ikizler TA. Nutrition in kidney disease: core curriculum 2022. *Am J Kidney Dis* 2022; **79:** 437-449.

90. Greenwood SA, Koufaki P, Rush R, *et al.* Exercise counselling practices for patients with chronic kidney disease in the UK: a renal multidisciplinary team perspective. *Nephron Clin Pract* 2014; **128:** 67-72.

91. Ikizler TA, Kramer HJ, Beddhu S, *et al.* ASN kidney health guidance on the management of obesity in persons living with kidney diseases. *J Am Soc Nephrol* 2024; **35:** 1574-1588.

92. Sattar N, Lee MMY, Kristensen SL, *et al.* Cardiovascular, mortality, and kidney outcomes with GLP-1 receptor agonists in patients with type 2 diabetes: a systematic review and meta-analysis of randomised trials. *Lancet Diabetes Endocrinol* 2021; **9:** 653-662.

93. Perkovic V, Tuttle KR, Rossing P, *et al.* Effects of semaglutide on chronic kidney disease in patients with type 2 diabetes. *N Engl J Med* 2024; **391:** 109-121.

94. Colhoun HM, Lingvay I, Brown PM, *et al.* Long-term kidney outcomes of semaglutide in obesity and cardiovascular disease in the SELECT trial. *Nat Med* 2024; **30:** 2058-2066.

95. Badve SV, Bilal A, Lee MMY, *et al.* Effects of GLP-1 receptor agonists on kidney and cardiovascular disease outcomes: a meta-analysis of randomised controlled trials. *Lancet Diabetes Endocrinol* 2025; **13:** 15-28.

96. Sahi SS, Garcia Valencia O, Na J, *et al.* Benefits of glucagon-like peptide-1 receptor agonists after kidney transplantation. *Endocr Pract* 2025**:** S1530-1891X(1525)00068-00060.

97. Orandi BJ, Chen Y, Li Y, *et al.* GLP-1 receptor agonists in kidney transplant recipients with pre-existing diabetes: A retrospective cohort study. *Lancet Diabetes Endocrinol* 2025; **13:** 374-383.

98. Weeda ER, Muraoka AK, Brock MD, *et al.* Medication adherence to injectable glucagon-like peptide-1 (GLP-1) receptor agonists dosed once weekly vs once daily in patients with type 2 diabetes: A meta-analysis. *Int J Clin Pract* 2021; **75:** e14060.

99. O'Hara DV, Janse RJ, Fu EL, *et al.* Adherence and persistence to novel glucose-lowering medications in persons with type 2 diabetes mellitus undergoing routine care. *Diabetes Res Clin Pract* 2024; **213:** 111745.

100. Weiss T, Carr RD, Pal S, *et al.* Real-world adherence and discontinuation of glucagon-like peptide-1 receptor agonists therapy in type 2 diabetes mellitus patients in the United States. *Patient Prefer Adherence* 2020; **14:** 2337-2345.

101. Hwang JH, Laiteerapong N, Huang ES, *et al.* Lifetime health effects and cost-effectiveness of tirzepatide and semaglutide in US adults. *JAMA Health Forum* 2025; **6:** e245586-e245586.

102. Colhoun HM, Kahn SE, Brown PM, *et al.* Is semaglutide as effective at reducing major cardiovascular events in the presence of impaired kidney function in people with overweight or obesity? A prespecified analysis from the SELECT trial. Abstract 146. Presented at the 2024 European Association for the Study of Diabetes Annual Meeting, Madrid, Spain. Available at: https://drive.google.com/file/d/11jdID0Hl7evB7bRn7ApwW3-Agf1NhTby/view. Accessed June 16, 2025.

103. Apperloo EM, Gorriz JL, Soler MJ, *et al.* Semaglutide in patients with overweight or obesity and chronic kidney disease without diabetes: a randomized double-blind placebo-controlled clinical trial. *Nat Med* 2025; **31:** 278-285.

104. Heerspink HJL, Sattar N, Pavo I, *et al.* Effects of tirzepatide versus insulin glargine on kidney outcomes in type 2 diabetes in the SURPASS-4 trial: Post-hoc analysis of an open-label, randomised, phase 3 trial. *Lancet Diabetes Endocrinol* 2022; **10:** 774-785.

105. Heerspink HJL, Sattar N, Pavo I, *et al.* Effects of tirzepatide versus insulin glargine on cystatin C-based kidney function: A SURPASS-4 post hoc analysis. *Diabetes Care* 2023; **46:** 1501-1506.

106. Apperloo EM, Tuttle KR, Pavo I, *et al.* Tirzepatide associated with reduced albuminuria in participants with type 2 diabetes: pooled post hoc analysis from the randomized active- and placebo-controlled SURPASS-1-5 Clinical Trials. *Diabetes Care* 2025; **48:** 430-436.

107. Selvarajah V, Robertson D, Hansen L, *et al.* A randomized phase 2b trial examined the effects of the glucagon-like peptide-1 and glucagon receptor agonist cotadutide on kidney outcomes in patients with diabetic kidney disease. *Kidney Int* 2024; **106:** 1170-1180.

108. Heerspink HL, Lu Z, Du Y, *et al.* Effect of retatrutide on kidney parameters in people with type 2 diabetes and/or obesity—a post-hoc analysis of two phase 2 trials. *Diabetes* 2024; **73 (Supplement 1):** Abstract 754-P.

109. Kunutsor SK, Zaccardi F, Balasubramanian VG, *et al.* Glycaemic control and macrovascular and microvascular outcomes in type 2 diabetes: Systematic review and meta-analysis of cardiovascular outcome trials of novel glucose-lowering agents. *Diabetes Obes Metab* 2024; **26:** 1837-1849.

110. MacLaughlin HL, Cook SA, Kariyawasam D, *et al.* Nonrandomized trial of weight loss with orlistat, nutrition education, diet, and exercise in obese patients with CKD: 2-year follow-up. *Am J Kidney Dis* 2010; **55:** 69-76.

111. Weir MA, Beyea MM, Gomes T, *et al.* Orlistat and acute kidney injury: An analysis of 953 patients. *Arch Intern Med* 2011; **171:** 703-704.

112. Mavrakanas TA, Tsoukas MA, Brophy JM, *et al.* SGLT-2 inhibitors improve cardiovascular and renal outcomes in patients with CKD: A systematic review and meta-analysis. *Sci Rep* 2023; **13:** 15922.

113. Cheong AJY, Teo YN, Teo YH, *et al.* SGLT inhibitors on weight and body mass: A meta-analysis of 116 randomized-controlled trials. *Obesity (Silver Spring)* 2022; **30:** 117-128.

114. Mann JFE, Rossing P, Bakris G, *et al.* Effects of semaglutide with and without concomitant SGLT2 inhibitor use in participants with type 2 diabetes and chronic kidney disease in the FLOW trial. *Nat Med* 2024; **30:** 2849-2856.

115. Weghuber D, Barrett T, Barrientos-Pérez M, *et al.* Once-weekly semaglutide in adolescents with obesity. *N Engl J Med* 2022; **387:** 2245-2257.

116. Mishra T, Shapiro JB, Ramirez L, *et al.* Nephrolithiasis after bariatric surgery: A comparison of laparoscopic Roux-en-Y gastric bypass and sleeve gastrectomy. *Am J Surg* 2020; **219:** 952-957.

117. Chang AR, Chen Y, Still C, *et al.* Bariatric surgery is associated with improvement in kidney outcomes. *Kidney Int* 2016; **90:** 164-171.

118. Alothman S, Cornejo J, Adrales G, *et al.* Comparative outcomes of bariatric surgery in patients with ESRD on dialysis in the modern era of renal transplantation: analysis using the 2015-2020 MBSAQIP database. *Surg Endosc* 2023; **37:** 7106-7113.

119. Aboueisha MA, Evans L, Allotey JK, *et al.* A 5-year propensity-matched analysis of perioperative outcomes in patients with chronic kidney disease undergoing bariatric surgery. *Surg Endosc* 2023; **37:** 2335-2346.

120. Pencovich N, Long JJ, Smith BH, *et al.* Outcomes of kidney transplantation in patients that underwent bariatric surgery: A systematic review and meta-analysis. *Transplantation* 2024; **108:** 346-356.

121. Wathanavasin W, Banjongjit A, Avihingsanon Y, *et al.* Prevalence of sarcopenia and its impact on cardiovascular events and mortality among dialysis patients: a systematic review and meta-analysis. *Nutrients* 2022; **14:** 4077.

122. Ikizler TA, Burrowes JD, Byham-Gray LD, *et al.* KDOQI Clinical Practice Guideline for Nutrition in CKD: 2020 Update. *Am J Kidney Dis* 2020; **76:** S1-s107.

123. Armstrong SC, Bolling CF, Michalsky MP, *et al.* Pediatric metabolic and bariatric surgery: evidence, barriers, and best practices. *Pediatrics* 2019; **144:** e20193223.

124. Nehus EJ, Khoury JC, Inge TH, *et al.* Kidney outcomes three years after bariatric surgery in severely obese adolescents. *Kidney Int* 2017; **91:** 451-458.

125. Cohen RV, Pereira TV, Aboud CM, *et al.* Gastric bypass versus best medical treatment for diabetic kidney disease: 5 years follow up of a single-centre open label randomised controlled trial. *EClinicalMedicine* 2022; **53:** 101725.

126. Aminian A, Gasoyan H, Zajichek A, *et al.* Renoprotective effects of metabolic surgery versus GLP1 receptor agonists on progression of kidney impairment in patients with established kidney disease. *Ann Surg* 2024; **280:** 414-423.

127. Wharton S, Lau DCW, Vallis M, *et al.* Obesity in adults: a clinical practice guideline. *Canadian Medical Association Journal* 2020; **192:** E875.

128. González AM, Gutman T, Lopez-Vargas P, *et al.* Patient and Caregiver Priorities for Outcomes in CKD: A Multinational Nominal Group Technique Study. *Am J Kidney Dis* 2020; **76:** 679-689.

129. ICHOM International Consortium for Health Outcomes Measurement: Patient-Centered Outcome Measures: Adult Obesity. In (vol 2025), edited by ICHOM International Consortium for Health Outcomes Measurement, 2024

130. Tang E, Yantsis A, Ho M, *et al.* Patient-Reported Outcome Measures for Patients With CKD: The Case for Patient-Reported Outcomes Measurement Information System (PROMIS) Tools. *Am J Kidney Dis* 2024; **83:** 508-518.

131. Ndumele CE, Neeland IJ, Tuttle KR, *et al.* A Synopsis of the Evidence for the Science and Clinical Management of Cardiovascular-Kidney-Metabolic (CKM) Syndrome: A Scientific Statement From the American Heart Association. *Circulation* 2023; **148:** 1636-1664.

132. Chew NWS, Mehta A, Goh RSJ, *et al.* Cardiovascular-liver-metabolic health: recommendations in screening, diagnosis, and management of metabolic dysfunction-associated steatotic liver disease in cardiovascular disease via modified Delphi approach. *Circulation* 2025; **151:** 98-119.

133. Chan JCN, Lim LL, Wareham NJ, *et al.* The Lancet Commission on diabetes: using data to transform diabetes care and patient lives. *Lancet* 2021; **396:** 2019-2082.

134. Lim LL, Lau ESH, Ozaki R, *et al.* Association of technologically assisted integrated care with clinical outcomes in type 2 diabetes in Hong Kong using the prospective JADE Program: A retrospective cohort analysis. *PLoS Med* 2020; **17:** e1003367.

135. Kelly JT, Jegatheesan DK, Dawson J, *et al.* Are Digital Health Technologies and Models of Nutrition Care the Future of Chronic Kidney Disease Management? *J Ren Nutr* 2023; **33:** S80-s87.

136. Tobias DK, Merino J, Ahmad A, *et al.* Second international consensus report on gaps and opportunities for the clinical translation of precision diabetes medicine. *Nat Med* 2023; **29:** 2438-2457.

137. Acosta A, Camilleri M, Abu Dayyeh B, *et al.* Selection of antiobesity medications based on phenotypes enhances weight loss: a pragmatic trial in an obesity clinic. *Obesity* 2021; **29:** 662-671.

138. KDIGO Conference Participants. Genetics in chronic kidney disease: conclusions from a Kidney Disease: Improving Global Outcomes (KDIGO) Controversies Conference. *Kidney Int* 2022; **101:** 1126-1141.

139. Vallis M, Piccinini-Vallis H, Sharma AM, *et al.* Clinical review: modified 5 As: minimal intervention for obesity counseling in primary care. *Can Fam Physician* 2013; **59:** 27-31.

140. Jensen SBK, Blond MB, Sandsdal RM, *et al.* Healthy weight loss maintenance with exercise, GLP-1 receptor agonist, or both combined followed by one year without treatment: a post-treatment analysis of a randomised placebo-controlled trial. *EClinicalMedicine* 2024; **69:** 102475.

141. Pfeffer MA, Claggett B, Diaz R, *et al.* Lixisenatide in patients with type 2 diabetes and acute coronary syndrome. *N Engl J Med* 2015; **373:** 2247-2257.

142. Marso SP, Daniels GH, Brown-Frandsen K, *et al.* Liraglutide and cardiovascular outcomes in type 2 diabetes. *N Engl J Med* 2016; **375:** 311-322.

143. Mann JFE, Ørsted DD, Brown-Frandsen K, *et al.* Liraglutide and renal outcomes in type 2 diabetes. *N Engl J Med* 2017; **377:** 839-848.

144. Marso SP, Bain SC, Consoli A, *et al.* Semaglutide and cardiovascular outcomes in patients with type 2 diabetes. *N Engl J Med* 2016; **375:** 1834-1844.

145. Holman RR, Bethel MA, Mentz RJ, *et al.* Effects of once-weekly exenatide on cardiovascular outcomes in type 2 diabetes. *N Engl J Med* 2017; **377:** 1228-1239.

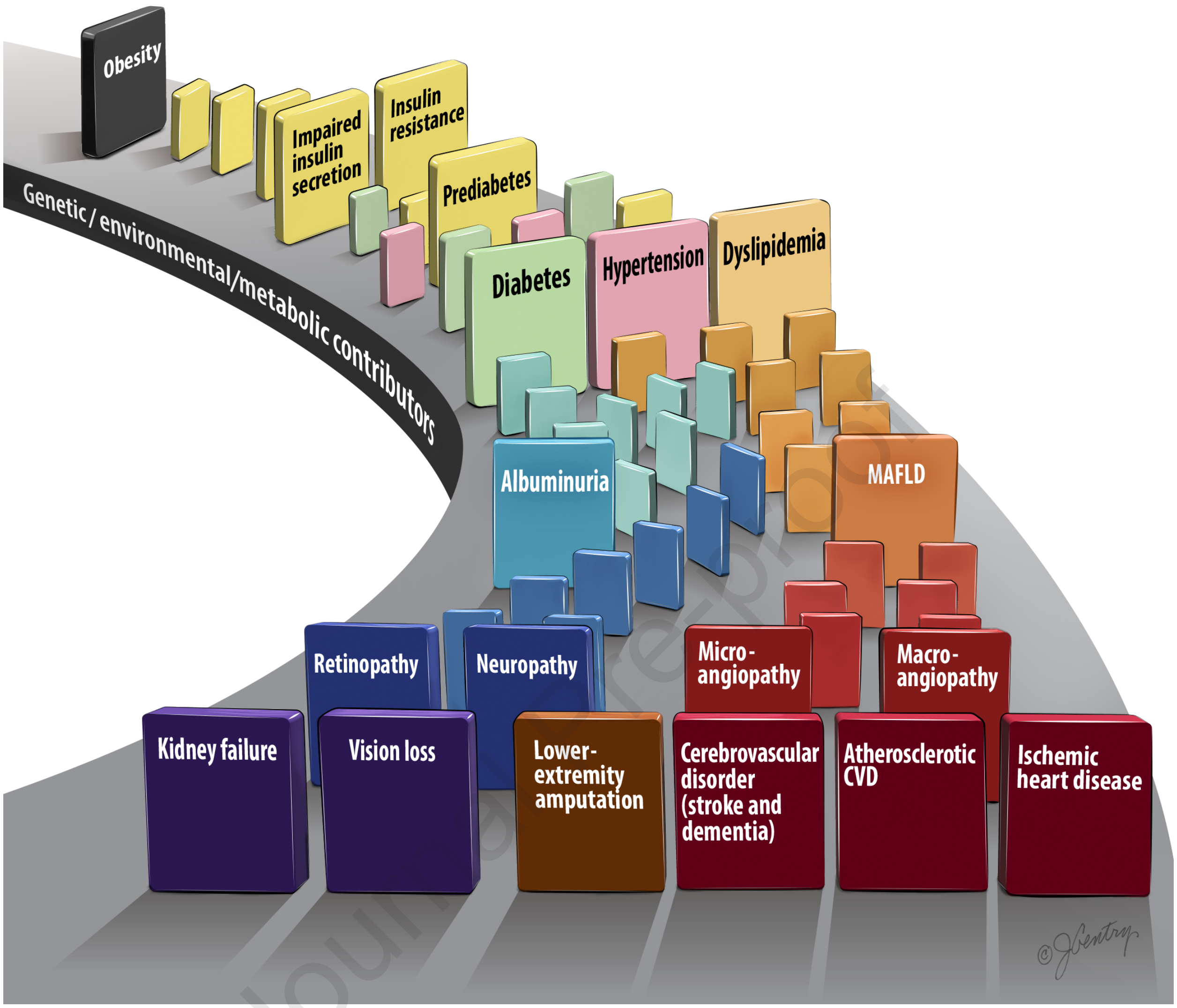
146. Gerstein HC, Colhoun HM, Dagenais GR, *et al.* Dulaglutide and cardiovascular outcomes in type 2 diabetes (REWIND): a double-blind, randomised placebo-controlled trial. *Lancet* 2019; **394:** 121-130.

147. Gerstein HC, Colhoun HM, Dagenais GR, *et al.* Dulaglutide and renal outcomes in type 2 diabetes: an exploratory analysis of the REWIND randomised, placebo-controlled trial. *Lancet* 2019; **394:** 131-138.

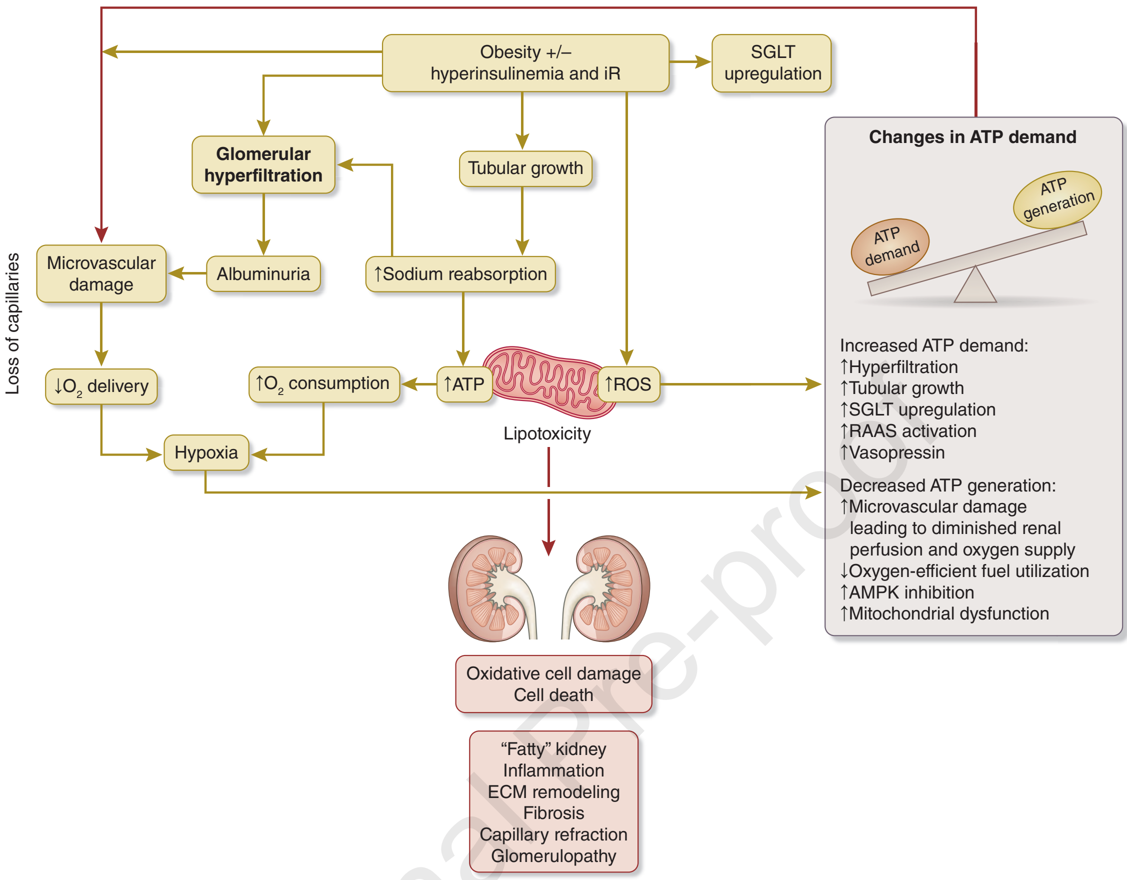
148. Hernandez AF, Green JB, Janmohamed S, *et al.* Albiglutide and cardiovascular outcomes in patients with type 2 diabetes and cardiovascular disease (Harmony Outcomes): a double-blind, randomised placebo-controlled trial. *Lancet* 2018; **392:** 1519-1529.

149. Husain M, Birkenfeld AL, Donsmark M, *et al.* Oral semaglutide and cardiovascular outcomes in patients with type 2 diabetes. *N Engl J Med* 2019; **381:** 841-851.

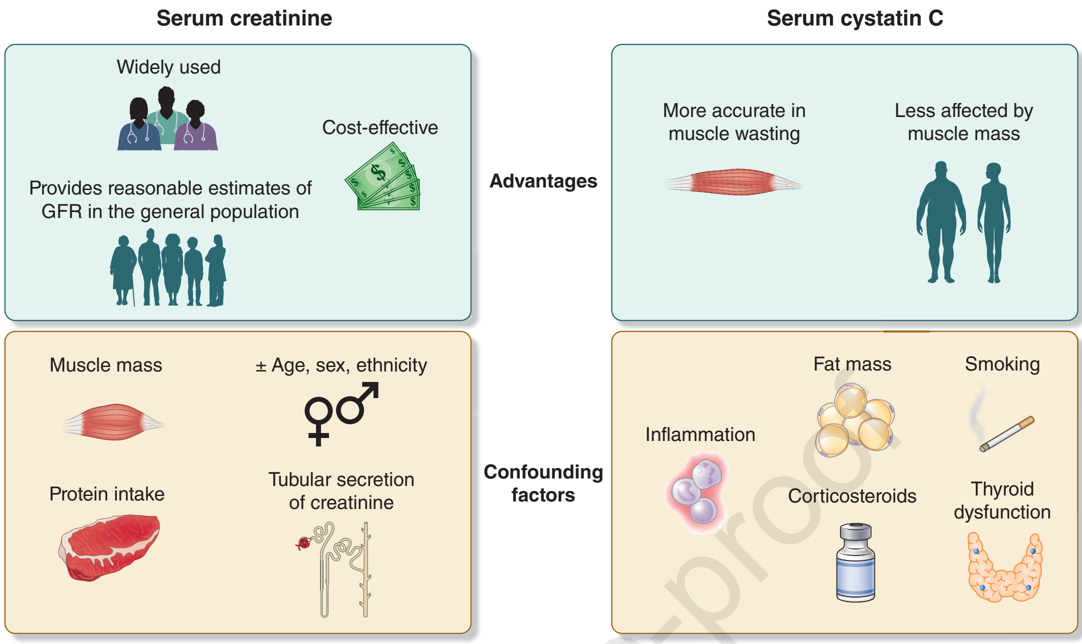
150. Gerstein HC, Sattar N, Rosenstock J, *et al.* Cardiovascular and renal outcomes with efpeglenatide in type 2 diabetes. *N Engl J Med* 2021; **385:** 896-907.

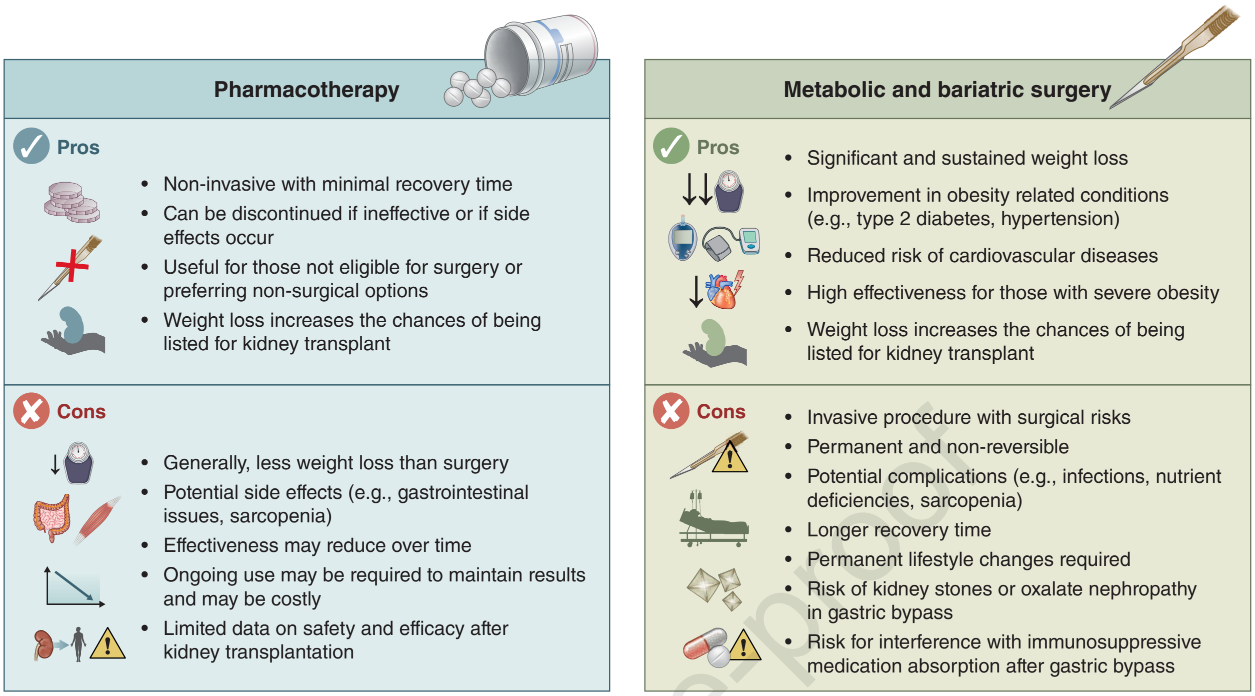


Obesity
Impaired insulin secretion
Insulin resistance
Prediabetes
Genetic / environmental/metabolic contributors
Diabetes
Hypertension
Dyslipidemia
Albuminuria
MAFLD
Retinopathy
Neuropathy
Micro-angiopathy
Macro-angiopathy
Kidney failure
Vision loss
Lower-extremity amputation
Cerebrovascular disorder (stroke and dementia)
Atherosclerotic CVD
Ischemic heart disease

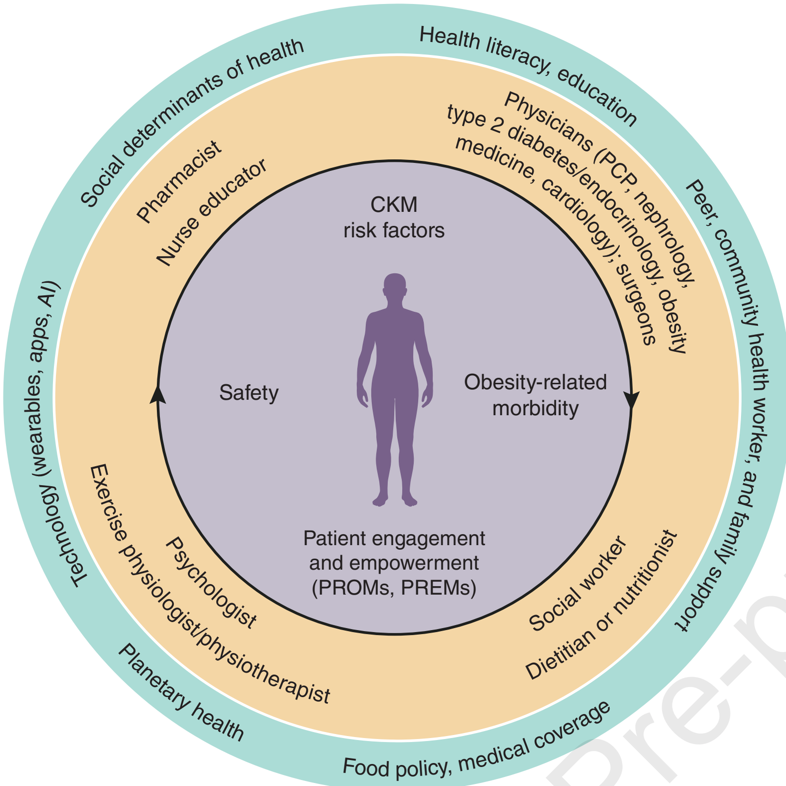


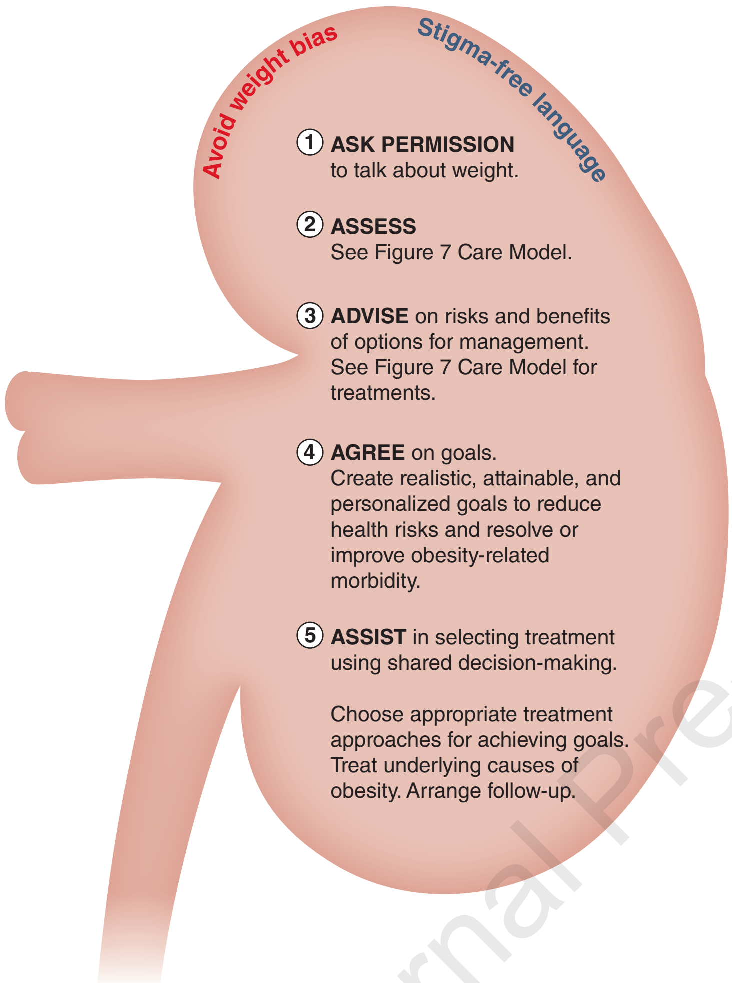
Obesity +/– hyperinsulinemia and iR
SGLT upregulation
Glomerular hyperfiltration
Tubular growth
Microvascular damage
Albuminuria
↑Sodium reabsorption
Loss of capillaries
↓O_2 delivery
↑O_2 consumption
↑ATP
↑ROS
Lipotoxicity
Hypoxia
Changes in ATP demand
ATP demand
ATP generation
Increased ATP demand:
↑Hyperfiltration
↑Tubular growth
↑SGLT upregulation
↑RAAS activation
↑Vasopressin
Decreased ATP generation:
↑Microvascular damage leading to diminished renal perfusion and oxygen supply
↓Oxygen-efficient fuel utilization
↑AMPK inhibition
↑Mitochondrial dysfunction
Oxidative cell damage
Cell death
"Fatty" kidney
Inflammation
ECM remodeling
Fibrosis
Capillary refraction
Glomerulopathy

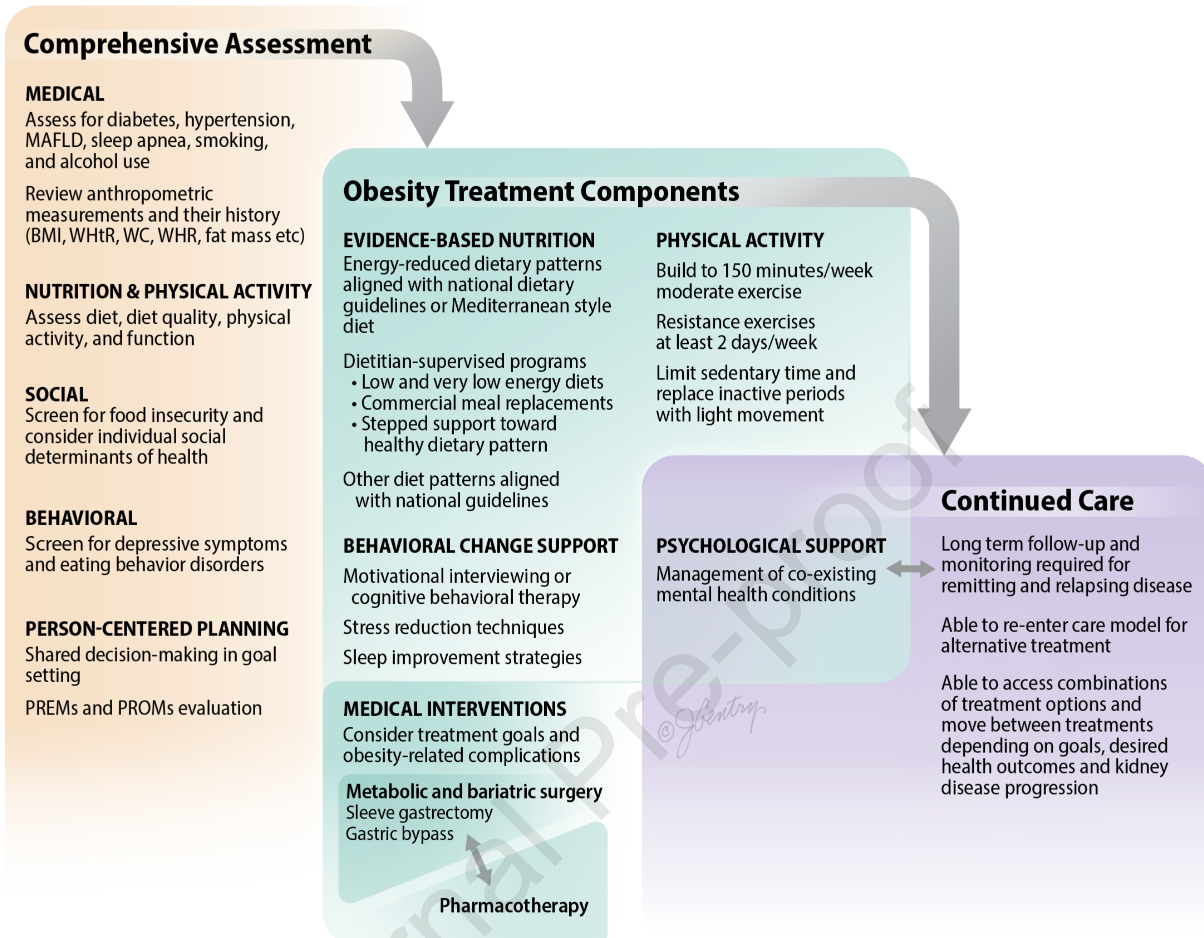

Serum creatinine
Serum cystatin C
Widely used
Cost-effective
Provides reasonable estimates of GFR in the general population
Advantages
More accurate in muscle wasting
Less affected by muscle mass
Muscle mass
± Age, sex, ethnicity
Protein intake
Tubular secretion of creatinine
Confounding factors
Fat mass
Smoking
Inflammation
Corticosteroids
Thyroid dysfunction



Pharmacotherapy
Pros
• Non-invasive with minimal recovery time
• Can be discontinued if ineffective or if side effects occur
• Useful for those not eligible for surgery or preferring non-surgical options
• Weight loss increases the chances of being listed for kidney transplant
Cons
• Generally, less weight loss than surgery
• Potential side effects (e.g., gastrointestinal issues, sarcopenia)
• Effectiveness may reduce over time
• Ongoing use may be required to maintain results and may be costly
• Limited data on safety and efficacy after kidney transplantation
Metabolic and bariatric surgery
Pros
• Significant and sustained weight loss
• Improvement in obesity related conditions (e.g., type 2 diabetes, hypertension)
• Reduced risk of cardiovascular diseases
• High effectiveness for those with severe obesity
• Weight loss increases the chances of being listed for kidney transplant
Cons
• Invasive procedure with surgical risks
• Permanent and non-reversible
• Potential complications (e.g., infections, nutrient deficiencies, sarcopenia)
• Longer recovery time
• Permanent lifestyle changes required
• Risk of kidney stones or oxalate nephropathy in gastric bypass
• Risk for interference with immunosuppressive medication absorption after gastric bypass


Health literacy, education
Peer, community health worker, and family support
Food policy, medical coverage
Planetary health
Technology (wearables, apps, AI)
Social determinants of health
Physicians (PCP, nephrology, type 2 diabetes/endocrinology, obesity medicine, cardiology); surgeons
Social worker
Dietitian or nutritionist
Psychologist
Exercise physiologist/physiotherapist
Pharmacist
Nurse educator
CKM risk factors
Obesity-related morbidity
Patient engagement and empowerment (PROMs, PREMs)
Safety


Avoid weight bias
Stigma-free language
1 ASK PERMISSION to talk about weight.
2 ASSESS See Figure 7 Care Model.
3 ADVISE on risks and benefits of options for management. See Figure 7 Care Model for treatments.
4 AGREE on goals. Create realistic, attainable, and personalized goals to reduce health risks and resolve or improve obesity-related morbidity.
5 ASSIST in selecting treatment using shared decision-making.
Choose appropriate treatment approaches for achieving goals. Treat underlying causes of obesity. Arrange follow-up.


Comprehensive Assessment
MEDICAL
Assess for diabetes, hypertension, MAFLD, sleep apnea, smoking, and alcohol use
Review anthropometric measurements and their history (BMI, WHtR, WC, WHR, fat mass etc)
NUTRITION & PHYSICAL ACTIVITY
Assess diet, diet quality, physical activity, and function
SOCIAL
Screen for food insecurity and consider individual social determinants of health
BEHAVIORAL
Screen for depressive symptoms and eating behavior disorders
PERSON-CENTERED PLANNING
Shared decision-making in goal setting
PREMs and PROMs evaluation
Obesity Treatment Components
EVIDENCE-BASED NUTRITION
Energy-reduced dietary patterns aligned with national dietary guidelines or Mediterranean style diet
Dietitian-supervised programs
• Low and very low energy diets
• Commercial meal replacements
• Stepped support toward healthy dietary pattern
Other diet patterns aligned with national guidelines
PHYSICAL ACTIVITY
Build to 150 minutes/week moderate exercise
Resistance exercises at least 2 days/week
Limit sedentary time and replace inactive periods with light movement
BEHAVIORAL CHANGE SUPPORT
Motivational interviewing or cognitive behavioral therapy
Stress reduction techniques
Sleep improvement strategies
PSYCHOLOGICAL SUPPORT
Management of co-existing mental health conditions
MEDICAL INTERVENTIONS
Consider treatment goals and obesity-related complications
Metabolic and bariatric surgery
Sleeve gastrectomy
Gastric bypass
Pharmacotherapy
Continued Care
Long term follow-up and monitoring required for remitting and relapsing disease
Able to re-enter care model for alternative treatment
Able to access combinations of treatment options and move between treatments depending on goals, desired health outcomes and kidney disease progression